

The Impact of Digital Transformation on Productivity: Evidence from Chinese Manufacturing Firms

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Abstract

This study examines why digital transformation (DT) may fail to generate immediate productivity gains, using evidence from Chinese manufacturing firms. Drawing on dynamic capability and institutional perspectives, we conceptualize DT as an organizational process whose performance effects depend on firms' ability to absorb, coordinate, and deploy new digital resources. Using a text-based measure of DT strategies and a staggered difference-in-differences design, we find that DT is associated with an average productivity decline of about 10% over six years. This finding challenges the view that DT automatically enhances productivity and contributes to research on the productivity paradox. We further show that the productivity effects of DT vary systematically with CEO tenure, firm size, firm age, and skill intensity, emphasizing the role of internal institutional conditions in shaping firms' adjustment and capability-building processes. Additionally, DT is associated with greater access to long-term bank loans, suggesting a financial signaling effect. This study advances management research on DT by highlighting its uneven operational and financial consequences in an emerging economies.

Keywords: digital transformation, productivity, staggered difference-in-differences, strategic management

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1 Introduction

The global economy is rapidly transitioning into a digital era, with digital technologies continuously reshaping various industries. Digital transformation (DT) has become a critical strategic priority, especially in the manufacturing sector, which is vital for China's economic growth and global competitiveness. DT in manufacturing involves integrating digital technologies into processes and products to enhance efficiency and quality (Furr et al., 2022). A substantial literature links information technology and digital investment to productivity and value creation (Brynjolfsson and Hitt, 1996, 2000; Bharadwaj et al., 2013). At the same time, other studies show that the benefits of DT depend on organizational adaptation, complementary capabilities, and implementation conditions (Vial, 2019; Hanelt et al., 2021; Simsek et al., 2024; Yoo et al., 2024). Despite its potential, the implementation of DT is fraught with challenges and often yields unsatisfactory outcomes.¹

To address the issue, we theorize DT not as a discrete technology adoption decision, but as a multi-stage organizational process whose consequences depend on capability formation, internal institutional conditions, and stakeholder evaluation. Drawing on dynamic capability theory, we contend that DT may trigger resource reconfiguration and adaptation efforts without generating immediate productivity payoffs (Warner and Wäger, 2019). Drawing on institutional theory, we further argue that the productivity effects of DT are conditioned by firms' internal normative, regulative, and culture-cognitive arrangements (Scott, 2001). In this study, these conditions are captured by CEO tenure, firm size, firm age, and skill intensity. We also consider an external dimension of DT by examining whether DT affects firms' access to finance through a signaling mechanism.

We examine these issues using a panel of Chinese listed manufacturing firms from 2010 to 2018. China provides an informative setting because manufacturing firms have faced strong pressure to digitalize while operating in an environment marked by substantial heterogeneity in organizational capability, digital readiness, and financing conditions. To capture DT as a strategic orientation rather than a one-time technology purchase, we construct a text-based measure from firms' annual reports and estimate its effects using a staggered difference-in-differences(DID) design.

Our analysis introduces novel measures of DT derived from text analysis of annual reports, capturing DT as a business strategy rather than focusing on specific technologies.² In detail, we perform text analysis on annual reports of Chinese publicly trade firms to detect firms' strategy of DT using keywords related to DT.³ Instead of focusing on specific digital technologies, our

¹The successful rate of digital transformation is found to be low. According to a global survey conducted by McKinsey on DT, in 2016, only 20% of the surveyed companies achieved success, compared with 26% in 2014 and 20% in 2012 (McKinsey, 2018). Whether DT actually enhances firm productivity therefore remains an open question.

²A set of digital technologies that triggered and shaped DT is the so-called SMACIT technologies (social, mobile, analytics, cloud, and Internet of Things) (Sebastian et al., 2020).

³An annual report is a document that public corporations must provide annually to shareholders that describes their operations and financial conditions.

measure of DT captures it as a business strategy, thus suiting our empirical context. Besides, the choice of DT is endogenous in the sense that the onset of digitalization is partly determined by the firm's own characteristics.⁴ This approach aligns with our empirical context and addresses the challenge of defining DT. Additionally, we incorporate unique indicators of managerial incentives and financial signaling to explore the underlying mechanisms of DT's impact.

Our findings reveal that DT, on average, decreases firm productivity by approximately 10%, increases total assets by 6%, and total employees by 8%, without significant improvement in total sales. We observe no significant structural changes in production and management organizations post-digitalization. The productivity decline is more pronounced for firms with short-tenured CEOs, smaller firms, younger firms, high-skill firms, and those in more competitive industries. Interestingly, firms that initiated DT later experience greater productivity drops.

Our contributions are threefold. First, our study advances the literature by examining the productivity paradox of DT within the context of Chinese manufacturing firms (Brynjolfsson and Hitt, 1996; Tambe and Hitt, 2012; Bloom et al., 2014). Our findings show that DT does not necessarily improve firm productivity. We refine dynamic capability theory by showing that the organizational and performance consequences of DT are more contingent and less automatic than is often assumed (Bavafa et al., 2018; Tan and Netessine, 2020). Methodologically, these findings are reinforced by our use of a staggered DID design, which strengthens causal identification.

Second, our study contributes to institutional perspectives by showing that the performance consequences of DT depend systematically on firms' internal institutional conditions. By linking CEO tenure, firm size, firm age, and skill intensity to firms' normative, regulative, and culture-cognitive arrangements, we explain why similar digital initiatives produce markedly different outcomes across firms. This contribution underscores that the effects of DT are contingent on the internal organizational environments through which digital initiatives are interpreted, coordinated, and absorbed.

Third, our study provides novel insights into the mechanisms driving firms to adopt DT despite observed productivity declines, contributing to the literature on strategic financial signaling in DT. We demonstrate that digitalized firms are more likely to secure increased bank loans, thereby alleviating financial constraints. This discovery suggests that beyond operational impacts, DT serves as a strategic tool for enhancing financial standing, particularly in particularly in environments characterized by information asymmetry and imperfect capital markets. This insight expands our understanding of the multifaceted benefits of DT.

2 Theoretical Framework and Hypotheses

⁴At summarized by Hanelt et al. (2021), organizational characteristics (*organizational strategy and legacy*) and managerial characteristics (*digitalization awareness of TMT*) can both affect the DT strategy.

2.1 Digital Transformation and the Role of Digital Strategies

DT signifies not merely the adoption of discrete technologies, but a firm-wide, strategic realignment of resources, capabilities, and value creation logics for competitive advantages (Dery et al., 2017; Teece, 2018; Plekhanov et al., 2023). Drawing on recent advances in digital business strategy (Bharadwaj et al., 2013; Sebastian et al., 2020; Siachou et al., 2021) and digital innovation (Nambisan et al., 2019), we propose that firms develop coherent digital strategies along three interrelated dimensions to achieve DT: (1) organizational configuration and core organizational activities; (2) processes and value-chain periphery; and (3) external environment and ecosystem orchestration. By articulating DT as a portfolio of strategic choices rather than a monolithic construct, we integrate dynamic-capabilities and institutional perspectives to explain heterogeneous productivity outcomes.

At its core, a digital strategy requires reconfiguring organizational structure and governance to enable rapid decision-making, seamless cross-functional coordination, and the full integration of digital capabilities (Hess et al., 2016). Firms must balance centralization—ensuring consistency in platform architectures and data standards—with decentralization, which empowers cross-functional teams to prototype solutions and adapt to local market signals (Sebastian et al., 2020). Establishing digital "hubs" or "labs", embedding chief digital officers, and deploying agile governance cells are strategic levers that shape how quickly and effectively firms sense digital opportunities, reallocate resources, and integrate new capabilities (Hanelt et al., 2021; Li et al., 2024).

Beyond internal design, digital strategies re-engineer core processes and extend to upstream and downstream partners (Porter et al., 2014; van Meeteren et al., 2022). Two archetypal approaches prevail. Incremental process digitalization substitutes legacy routines with automation, advanced analytics, and IoT sensors to optimize cost and quality (Cheng et al., 2023; Zhang et al., 2023). In addition, platform-enabled co-creation transforms the firm's role from producer to orchestrator by integrating suppliers, customers, and innovators via digital marketplaces, APIs, and shared data standards (Parker et al., 2016). For example, blockchain-backed traceability and real-time logistics coordination shift the locus of control in the supply chain and unlock new service revenues (Adner, 2017).

Finally, DT encompasses a firm's positioning and governance within broader value ecosystems (Adner, 2017; Jacobides et al., 2018). Keystone firms deploy platform orchestration tactics, setting governance rules, partner certification, and revenue-sharing models, to harness external innovation and generate network effects (Eaton et al., 2015; Priem et al., 2012). Alternatively, niche leaders pursue cooperative alliances, leveraging shared digital infrastructures to access strategic resources and lower entry barriers (Autio et al., 2018; Chatterjee et al., 2023). The efficacy of ecosystem orchestration is moderated by the firm's strategic resource endowments—data repositories, cloud infrastructures, and intellectual property—and by institutional contexts such as industry standards and regulatory regimes (Chen et al., 2022).

However, the shift towards DT introduces notable challenges that must be carefully man-

aged. The reliance on digital systems increases cybersecurity vulnerabilities, such as data breaches and hacking, which can compromise sensitive information and disrupt operations (Banker and Feng, 2019; He et al., 2020). Furthermore, the complexity of IT infrastructure necessitates robust security measures and continuous updates, which can be costly and resource-intensive (Bharadwaj et al., 2013). Organizational structures may become more complex, leading to misalignments between existing capabilities and new technological demands (Munch et al., 2018; Brynjolfsson et al., 2021). The introduction of new technologies often disrupts existing workflows, requiring extensive employee re-training and potentially resulting in temporary productivity losses. Additionally, firms may encounter substantial adjustment costs, both financial and operational, straining resources and leading to short-term inefficiencies (Goldfarb and Tucker, 2019; Feigenbaum and Gross, 2021). To effectively navigate these challenges, a strategic approach is essential, focusing on minimizing disruptions while maximizing the long-term benefits of DT.

2.2 Digital Transformation and Firm Productivity

As discussed above, DT represents a comprehensive integration of digital technologies across all facets of a business, fundamentally reshaping how firms operate and deliver values (Westerman et al., 2014; Liere-Netheler et al., 2018; Nambisan et al., 2019). This transformation is pivotal for maintaining competitiveness in the digital economy (Gong and Ribiere, 2021; Wessel et al., 2021). Nevertheless, the initial implementation of DT can pose significant challenges, potentially leading to sudden declines in productivity (Goldfarb and Tucker, 2019).

The potential of DT lies in its ability to enhance operational efficiency, drive innovation, and orchestrate external ecosystems (Kane et al., 2015; Furr et al., 2022). However, the pathway to realizing these benefits is fraught with complexity. The early stages of digital transformation often necessitate substantial investments in new technologies and the development of new organizational capabilities. These requirements can strain existing resources and disrupt established workflows, resulting in temporary inefficiencies (Helfat and Peteraf, 2003; Teece, 2018; Hanelt et al., 2021).

Empirical evidence underscores that while the adoption of digital technologies is necessary, it is insufficient on its own to guarantee immediate productivity gains (Brynjolfsson and Hitt, 1996; Tambe and Hitt, 2012; Brynjolfsson et al., 2021). The complexities associated with aligning digital technologies with existing processes can lead to compatibility issues, necessitating significant organizational changes and employee retraining (Eisenhardt and Martin, 2000; Feigenbaum and Gross, 2021). The time and effort required to reconfigure business strategies and develop dynamic capabilities are resource-intensive, potentially leading to a short-term decline in productivity (Teece et al., 1997; Brynjolfsson and McAfee, 2014).

A body of research on technological change and productivity has long recognized the potential for short-term disruptions during the adoption of new technologies (Miller and Tucker, 2011; Brynjolfsson et al., 2021). These disruptions are exacerbated by the need for firms to

adapt to rapidly changing technological landscapes, which often involves overcoming structural challenges and aligning new technologies with organizational cultures (Lawrence, 2020; Hanelt et al., 2021). The strategic imperative for DT is accompanied by substantial adjustment costs and organizational frictions. The complexity inherent in implementing digital technologies can lead to initial productivity declines as firms navigate these challenges and work to develop the necessary capabilities to fully leverage digital tools (Siggelkow and Levinthal, 2003; Feigenbaum and Gross, 2021). Due to the resource-intensive nature of capability development and the organizational disruptions associated with integrating new digital technologies, we propose that:

H1a: In the short run, DT is negatively associated with firm productivity.

In the long run, the strategic integration of digital technologies holds considerable promise for enhancing firm productivity. This period is characterized by the gradual alignment of DT initiatives with a firm's strategic objectives, which facilitates the realization of substantial productivity gains. Unlike the short run, where the immediate impacts of DT can be disruptive due to initial implementation challenges, the long-term perspective allows firms to fully leverage the potential of digital technologies as they mature within the organizational framework (Fernandez-Vidal et al., 2022; Oludapo et al., 2024).

Over time, DT enables firms to achieve enhanced operational efficiency, foster innovation, and improve customer engagement (Kane et al., 2015; Furr et al., 2022). This is achieved through the embedding of digital capabilities into the core operational processes, allowing firms to streamline operations and minimize inefficiencies. Such strategic integration creates a robust foundation for sustained growth and competitiveness in the digital economy. Firms can thus adapt more readily to evolving market conditions and technological advancements, as digital tools become seamlessly integrated into their strategic fabric (Pihir et al., 2019).

Prior study emphasizes that the mere adoption of digital technologies does not automatically translate into immediate productivity gains. Instead, it is the strategic alignment of these technologies with organizational goals over time that fosters a conducive environment for growth (Kretschmer and Khashabi, 2020; Sebastian et al., 2020). The transition involves overcoming initial compatibility issues and reconfiguring existing business processes. Once these challenges are addressed, firms can achieve significant long-term productivity improvements (Teece et al., 1997; Feigenbaum and Gross, 2021). As firms build the requisite capabilities and effectively align digital tools with their strategic objectives, they are poised to experience a gradual yet enduring enhancement in productivity (Lawrence, 2020; Hanelt et al., 2021). This process reflects the dynamic capability framework, where firms develop the ability to integrate and reconfigure resources in response to shifting environments (Teece, 2018). Hence, we posit that:

H1b: *In the long run, DT is positively associated with firm productivity.*

2.3 The Role of Institutional Contexts

The short-term impact of DT on firm productivity may be intricately linked to institutional conditions under which digital initiatives are implemented. If our hypothesis **H1a** is supported, it will require a deeper exploration of how DT should be theorized in the context, whether primarily as a process of formal organizational redesign or, more broadly, as a disruptive process of capability building and operational realignment.

An important possibility is that DT affects productivity before it produces observable changes in formal organizational structure. Existing research suggests that digital technologies often require firms to redesign workflows, decision processes, and coordination patterns (Kretschmer and Khashabi, 2020; Singh et al., 2020). At the same time, formal organizational structures are often sticky, especially in established firms characterized by legacy systems, entrenched routines, and resistance to change (Li, 2020). Therefore, firms may experience substantial implementation costs, learning frictions, and temporary operational disruption even when formal production and administrative structures do not change immediately. In this sense, the short-run productivity penalty of DT need not depend on visible structural reorganization alone. It may reflect incomplete capability conversion and difficulties in aligning digital initiatives with existing routines. We therefore suggest that DT may not be associated with significant short-term changes in formal organizational structures.

H2: *In the short run, DT does not lead to significant changes in organizational structures.*

Beyond formal structures, the productivity effects of DT are likely to vary systematically with firms' internal institutional conditions. Institutional theory suggests that organizational outcomes are shaped by internal normative, regulative, and culture-cognitive arrangements that influence how change is interpreted, supported, and absorbed (Lewin et al., 1999; Scott, 2001; Kogut et al., 2002). In the context of DT, we use CEO tenure, firm size, firm age, and skill intensity as theoretically grounded indicators of these internal conditions. Specifically, CEO tenure maps onto the normative pillar, firm size onto the regulative pillar, and firm age and skill intensity onto the culture-cognitive pillar. These attributes capture executive socialization and legitimacy, governance capacity and formalization, accumulated organizational experience, and workforce knowledge structure, thereby shaping whether digital initiatives can be effectively coordinated, legitimized, and translated into firm performance outcomes (Bharadwaj et al., 2013; Vial, 2019).

Within Scott's normative pillar, which concerns legitimacy, role expectations, and shared standards, CEO tenure shapes the authority and social embeddedness needed to orchestrate DT. Longer-tenured CEOs possess more firm-specific knowledge, stronger internal legitimacy, and a better understanding of historical routines and strategic priorities (Wang et al., 2016; Singh et al., 2020). These advantages make them better positioned to align stakeholders, reallocate resources, and integrate digital initiatives into legacy operations with less disruption (Helfat and Martin, 2015; Firk et al., 2021). By contrast, short-tenured CEOs often face weaker legitimacy and stronger pressure to deliver rapid results, while lacking deep firm-specific

knowledge (Hill and Phan, 1991; Lavery, 1996; Nayak et al., 2020; Chang et al., 2010). This combination makes DT implementation more fragmented.

The regulative pillar emphasizes the formal rules, governance structures, and compliance mechanisms that shape organizational behaviors and outcomes (Scott, 2001). While smaller firms benefit from minimal bureaucratic inertia, which facilitates rapid process realignment around novel digital tools, they simultaneously suffer from tighter resource constraints, weaker formal coordination routines, and less organizational slack (Damanpour, 1992; Laforet, 2013). These limitations reduce their ability to absorb the operational disruptions associated with early-stage DT. Larger firms, despite greater structural complexity, are more likely to possess formalized procedures, cross-functional expertise, and governance capacity that help cushion the productivity costs of implementation (Warner and Wäger, 2019; Sebastian et al., 2020). Thus, we argue that the negative productivity effect of DT to be stronger in smaller firms.

The culture-cognitive pillar (Levitt and March, 1988; Scott, 2001) highlights the role of shared understandings, organizational memory, and established interpretive schemas in shaping responses to change. Older firms possess more accumulated routines, interpretive frameworks, and training infrastructures that can help stabilize implementation and reduce the disorder associated with DT (Leonard-Barton, 1992; Helfat and Peteraf, 2003). On the contrary, younger firms are more likely to rely on emergent coordination, trial-and-error learning, and less institutionalized routines, all of which can magnify the operational disruption of digital initiatives (Santos and Eisenhardt, 2009; Warner and Wäger, 2019; Sebastian et al., 2020). Hence, we propose that the adverse productivity impact of DT is more pronounced in younger firms.

Under the culture-cognitive pillar of institutional theory, skill intensity reflects the degree to which tacit knowledge and professional mental models govern organizational activity (Orlikowski and Hofman, 1997; Scott, 2001). In firms with high skill intensity, employees often rely on professional routines developed through extensive training and repeated practice (Zammuto et al., 2007; Edmondson and Nembhard, 2009). Although these routines support quality and consistency, they can also make digital codification and automation more disruptive, especially when DT challenges established cognitive scripts and professional identities (Leonardi, 2012). Aligning such routines with new digital systems often involves steeper learning curves and higher adjustment costs (Lyytinen et al., 2016; Wessel et al., 2021). However, firms with lower skill intensity operate with fewer taken-for-granted assumptions, making them more receptive to digital tools and less prone to productivity drag. Hence, we propose that:

H3a: The negative relationship between DT and firm productivity is intensified in firms with shorter CEO tenures.

H3b: The negative relationship between DT and firm productivity is intensified in firms with smaller size.

H3c: The negative relationship between DT and firm productivity is intensified in younger firms.

H3d: The negative relationship between DT and firm productivity is intensified in firms with higher skill intensity.

2.4 Financial Signaling Effects of DT

DT invariably involves substantial investments in information technologies, process redesign, and workforce retraining, often resulting in an initial productivity decline (Brynjolfsson and McElheran, 2016; Nambisan et al., 2019). Yet, firms, particularly those in emerging markets marked by underdeveloped capital markets and acute information asymmetries (Allen et al., 2005; Beck et al., 2005; Merín-Rodríguez et al., 2024), persist in committing to DT. We contend that this persistence reflects DT’s role as a strategic financial signaling mechanism that alleviates capital constraints and secures improved financing conditions.

First, public announcements of large-scale digital initiatives (e.g., ERP deployments, IoT-enabled production lines) serve to mitigate information asymmetries between firms and lenders (Khanna, 2010; Liu et al., 2021). By signaling unobservable managerial foresight and technological competence, such disclosures reduce perceived default risks. Second, the articulation of a coherent digital roadmap enhances a firm’s legitimacy and reputation in the eyes of financial stakeholders (Huang et al., 2017). In capital-scarce environments, alignment with international best practices and institutional expectations confers reputational benefits that translate into higher credit ratings and broader access to debt financing. Third, DT investments that demonstrate rapid resource-process reconfiguration function as observable proxies for dynamic capabilities. Lenders and equity investors interpret these signals as indicators of a firm’s ability to adapt to technological change and sustain competitive advantage (Bolton et al., 2011; Guariglia and Yang, 2016). This strategic signaling is critical in mitigating financial constraints and enhancing access to capital (Davis and DeWitt, 2021), thereby enabling firms to navigate the challenges of DT with greater ease.

H4: The adoption of DT strategies leads to the acquisition of long-term and credit-based bank loans, as DT acts as a signaling mechanism that enhances a firm’s attractiveness to financial institutions.

3 Data

3.1 Research Design

China constitutes the world’s largest manufacturing economy, accounting for 30.2% of global manufacturing value-added in 2022.⁵ Despite its scale, the manufacturing sector exhibits substantial heterogeneity in digital adoption and continues to lag behind advanced economies in

⁵It is calculated based on publicly available World Bank data (<https://data.worldbank.org/indicator/NV.IND.MANF.CD>) and is also reported by the National Bureau of Statistics of China in the “Report on the 75th Anniversary of Economic and Social Development in New China (Series III)”.

digital integration (CAICT, 2021). This context enables us to systematically examine the dynamic impact of DT on firm productivity within the distinctive institutional environment and industrial diversity of an emerging economy. The empirical design of our study leverages a comprehensive dataset of Chinese manufacturing firms from 2010 to 2018, incorporating both firm-level and industry-level variables to rigorously assess the impact of DT on productivity. Critically, this window also marks a formative stage for the DT of Chinese manufacturing firms. The release of the “Twelfth Five-Year Plan for National Economic and Social Development” in 2011 initiated a national push toward industrial upgrading through information technology, while the launch of the “Internet Plus” action plan and “Made in China 2025” in 2015 further accelerated digital adoption and the integration of smart manufacturing technologies. Consistent with this trend, in our sample, the number of firms implementing DT increased substantially from 33 in 2010 to 170 in 2018. Therefore, the dataset captures a critical juncture when many Chinese firms began adopting DT as a strategic priority, supported by national development policies and advanced communication networks.

We employ a DID approach, exploiting the staggered adoption of DT across firms and industries over time. This method allows us to isolate the causal effects of DT from other contemporaneous changes, as well as dealing with the endogeneity issue caused by firm’s self-selection into DT (Callaway and Sant’Anna, 2021). Ideally, one would compare the performance of a digitalized firm with a non-digitalized firm that shares identical characteristics. However, observing the counterfactual performance for the DT mode not chosen by the firm is infeasible. The conventional DID techniques widely used in management studies may introduce biases when evaluating the impact of ICT or digital technologies (e.g., Pierce et al., 2015; Staats et al., 2017; Lu and Lu, 2017; Tan and Netessine, 2020). To address this issue, we innovatively adopt the recently developed estimator by Callaway and Sant’Anna (2021), which provides a semi-parametric method for constructing counterfactual performances using propensity-score-weighted outcomes of control units within a staggered DID framework.

3.2 Sources

We use data on publicly traded manufacturing firms in China. A-share market between 2010 and 2018. The period 2010 to 2018 coincides with a phase of consistent economic growth in China, avoiding the confounding effects of the 2008 global financial crisis and the 2020 pandemic (Hennig et al., 2023). We first scrap 15,994 annual reports for listed manufacturing firms from CNINF⁶. We obtain a rich set of balance-sheet variables from websites of Shanghai Stock Exchange, Shenzhen Stock Exchange, and CNINF. These annual reports allow us to detect firms’ digitalization strategies using text analysis. To complement the firm-level analysis, we obtain rich information from income statement, balance sheet, and cash flow statement

⁶CNINF (<http://www.cninfo.com.cn/new/index>) is the first large-scale professional website of securities in China to fully disclose the announcement information and market data of more than 2500 listed companies in Shenzhen and Shanghai.

from China Stock Market & Accounting Research (CSMAR) and employees' education from Wind Economic Database. This data construction approach allows us to rigorously capture the timing, extent, and heterogeneity of DT among Chinese manufacturing firms, and to robustly analyze its impact on productivity within the observed period.

3.3 Variables

For the purpose of empirical analysis, we construct several variables based on the firm-level information. We also use a bunch of firm-level variables to facilitate our analysis in controlling firm-level characteristics and uncovering underlying mechanisms.

3.3.1 Measuring Digital Transformation

DT is a complex and multifaceted concept that extends beyond the implementation of ICT-related technologies. While ICT tools are integral components of digitalization, they do not fully encapsulate its essence. We emphasize that the defining characteristic of DT lies in the strategic reform and comprehensive reconfiguration of business models, processes, and orientations driven by digital technologies. This holistic approach integrates digital tools into the core of business operations, reflecting deeper organizational change rather than isolated technological adoption.

To better capture the strategic content in DT, we rely on text analysis of annual reports to identify firms that have adopted DT. Annual reports, regulated by the Securities Supervision Commission, provide a comprehensive overview of a company's situation, market perspectives, and strategic priorities, making them a reliable source for assessing DT. Nonetheless, these reports are not without limitations. Firms may engage in "cheap talk," claiming digital strategies without providing substantive evidence or clear implementation details. To address this concern, we design a two-step procedure to identify firms that have adopted DT:

1. *Keyword Identification:* We first search for the keyword "digitalization" in the annual reports of publicly listed firms between 2010 and 2018. For each instance, we extract 200 words (in Chinese characters) before and after the keyword. Initial analysis reveals that DT primarily occurs in areas such as production, management, procurement, logistics, marketing, research and development, and digital products. However, we recognize that some firms may not explicitly use the term "digitalization" when describing strategic transformations facilitated by digital technologies. To capture a broader scope of digitalization, we construct a comprehensive list of keywords informed by empirical studies on DT. Examples include phrases such as "smart logistics," "robot-centric intelligent factories," "big data products," and "integrated management platforms." These keywords are used to extract additional relevant text from annual reports, following the same procedure as in the first step.

2. *Manual Validation:* To ensure the validity of our measure, we manually review the extracted texts to evaluate their relevance to a firm's own digitalization efforts. Texts describing industry- or country-level technological trends, which do not reflect firm-specific adoption, are

excluded. Following Zhai et al. (2022), two research assistants independently review the texts, and discrepancies are resolved through discussion and consensus. Additionally, firms that merely shift their business scope toward digital products without significant changes in operational processes are excluded. As a robustness check, we include cases where digital products may enable new business models or market penetration. Further details are provided in the data appendix.

Indeed, as with any text-based measure, our approach is not immune to the possibility that firms might overstate or prematurely articulate their digital strategies in annual reports to enhance investor appeal or signal future intentions rather than reflecting their current operational state. While this limitation is acknowledged, we argue that the use of such textual data is still valuable for capturing the emphasis firms place on DT as part of their strategic priorities. Additionally, we mitigate this concern by showing significant and positive responses in tangible and intangible asset investment in the data appendix. We also note that annual reports are critical communication tools governed by regulatory standards, which imposes some discipline on their content. Nevertheless, we acknowledge that our measure may reflect firms' aspirations or intentions rather than solely their realized digitalization, and we are careful to interpret our results within this context.

A variable *DT* is defined to summarize the information obtained by analyzing the digitalization-related texts in the listed firms' annual reports. We treat this variable as a firm-level policy variable. If the extracted texts are found to refer to DT, we define the *DT* to be one in the current year and afterwards. Otherwise, this variable is equal to zero. To perform meaningful analysis, we restrict that firms appear in the sample for at least two consecutive years.

Table 1: Trend of Digital Transformation in the Sample

Year	Number of Treated Firms	Treated Observations	Never Treated Observations
2010	33	400	884
2011	24	466	1009
2012	37	508	1067
2013	24	521	1079
2014	53	541	1143
2015	83	581	1216
2016	104	641	1323
2017	181	702	1511
2018	170	694	1490
Total	709	5054	10722

Note: The number of treated firms refers to the number of firms that started to adopt digital transformation in the corresponding year. Treated observations are the number of observations of corresponding cohort of treated firms. Never treated observations are the number of observations of firms that have never adopted digital transformation.

Table 1 reports the trend of DT. The number of treated firms increased from 33 in 2010 to 170 in 2018. During the sample period, 709 firms implemented DT. The treated group of firms

has a total of 5054 observations. The number of never-treated observations increased from 884 in 2010 to 1,490 in 2018, with a total of 10,722 observations.

We exclude firms that reported digitalization in the initial sample period because of a lack of pre-trend, and in the last sample period due to a lack of post-trend. This leads to a sample of 506 treated firms (3,695 observations) and 1,666 never treated firms (10,722 observations). The staggered rollout of DT motivates us to evaluate the impact of digitalization using the staggered DID estimator.

3.3.2 Firm Performance and Characteristics

Productivity. One of the core dependent variables we use is the firm-level total factor productivity (TFP). Productivity is one of the most popular indicators in existing studies on ICT (see Brynjolfsson and Hitt (2003), DeStefano et al. (2018), Tan and Netessine (2020), among others). Due to the lacking of price data, we use a revenue-based measure of TFP. In particular, we consider a logged revenue production function in Cobb-Douglas form:

$$vad_{ijt} = \beta_0 + \beta_k k_{ijt} + \beta_l l_{ijt} + TFP_{ijt} + \varepsilon_{ijt}, \quad (1)$$

where vad_{ijt} , k_{ijt} and l_{ijt} refer to firm i 's logged value added, capital stock, and labor in operating in industry j in year t , respectively. The variable TFP_{ijt} is the firm's TFP and ε_{ijt} is the exogenous shock to production. Because of the simultaneity bias, the ordinary least square (OLS) estimates of β_k and β_l for specification (1) are likely to be inconsistent and therefore bias the productivity estimates. The simultaneity issue arises because the productivity is correlated with input choices as firms make input decisions based on their own productivity. For example, more productive firms tend to invest more in capital and hire more workers. To address the problem of simultaneity bias, we follow Levinsohn and Petrin (2003) to use intermediate inputs to control for the unobserved productivity shocks and estimate the production function. Considering the industrial heterogeneity, we estimate the production function specification (1) for each industry, separately.⁷ We note that the productivity estimates contain information on firm's physical production efficiency and output's prices (Foster et al., 2008). Therefore, the productivity measure is more suitably called revenue TFP. The revenue-based TFP can also be understood as operational efficiency as named by (Malikov et al., 2021). Throughout our discussion, we do not make a distinction between revenue productivity and physical productivity, and treat the TFP measure as an integrated indicator for firm's overall efficiency.

To account for the possibility that digitalization may change the production technology of the firm, we also use labor productivity and capital efficiency to measure the production efficiency. Labor productivity is measure as the logged ratio between total sales and total employees, and capital efficiency is measured as the total sales divided by total fixed assets.

⁷We face data constraints as some industries have no enough observations. Therefore, we merge some industries together based on their similarity in main products. See our data appendix for details.

Firm Size. We also evaluate the change in firm size after digitalization. We employ three measures of firm size: logged total sales ($\ln(sales)$), logged total number of employees ($\ln(employees)$), and logged total assets ($\ln(assets)$). These measures are widely used in the ICT literature (DeStefano et al., 2018; Tan and Netessine, 2020). Total sales reflects information about the firm's products' prices and quantities, and total number of employees measure the total labor inputs. Total assets is the total amount of assets owned by the firm. This indicator is widely used in financial studies.

Production and Management Organizational Structures. We also consider the impact of DT on the production and organization structures. We use the capital intensity (kl) to measure the production's organization. The capital intensity is defined as the total fixed assets divided by total employees. The capital intensity measures the relative importance of capital in production activities. A higher capital intensity implies that firms rely more on capital in producing goods. We also consider the skill intensity ($skill_int$) of the firm. Following the literature on labor studies, we use the share of high-skilled workers (workers with bachelor's and higher degrees) in total employees as a measure of the firm's skill intensity. The technological upgrading incurred by adoption of ICT and/or digital technologies may be realized through disproportionately investing more intangibles assets (Brynjolfsson et al., 2021). Considering this, we also examine the impact of digitalization on the firm's intangibility ($intang_ratio$), which is defined as the ration between intangibles assets over the sum of intangibles and fixed assets (Property, Plant & Equipment, PP&E).

We measure the firm's management organizations using two alternative indicators. First, we use the share of administration workers in total employees ($admin_ratio$). Second, we also use the ratio of administration fees to total revenue ($admin_int$). These variable both reflect the relative importance of administration workers to production workers. Lower values of these variables may imply that firms manage the production workers through flat organization structures.

Other Firm Characteristics and Financial Variables. In our empirical analysis, we also use a series of variables to control firms' characteristics or as moderators. *CEO tenure* is measured as the number of years the CEO has served in their position, which captures the CEO's firm-specific experience, power base within the organization, and potential influence on DT initiatives and effectiveness. For relatively exogenous firm characteristics, we include age (age) and leverage ratios ($leverage$). In terms of the structure of corporate governance, we use three variables. First, we define a dummy variable *duality* that is equal to one if CEO is also the general manager. Second, we define a variable capturing the equity concentration measured by the shareholding ratio of the largest shareholder ($largestshare$). The third variable is the state-owned shareholding (soe), which is defined as the proportion of state-owned shares in total shares. To analyze the firm's financial motives for digitalization, we use information on firm's investment ($investment/assets$) and cash flow ($cash_flow/assets$). In addition, we employ detailed information on bank loans. Based on the loan's length of maturity, we consider long-term bank

loans (*long_loan*), short-term bank loans (*short_loan*). Based on the types of collaterals used for borrowing, we use variables of guaranteed loans (*guarant_loan*), credit loans (*credit_loan*), mortgage loans (*mortgage_loan*).⁸

3.3.3 Summary Statistics

To avoid the estimation bias caused by outliers, we exclude firms whose sales revenues are smaller than intermediate inputs. We trim the top 1% and bottom 1% of variables if possible outliers are found, and keep firms that appear at least two consecutive years. Moreover, we restrict our sample to be firms starting digitalization between year 2011 and 2017 to ensure we observe at least one year before and after the DT for the treated units. We provide a table summarizing the descriptive statistics of main variables in Table 2.

Table 2: Summary Statistics of Main Variables

Variables	N	mean	p50	sd	min	max
<i>DT</i>	14417	0.128	0	0.334	0	1
<i>TFP</i>	13368	5.372	5.06	1.779	1.688	8.785
<i>labor Eff.</i>	14400	13.581	13.509	0.801	5.825	18.771
<i>capital Eff.</i>	13382	0.56	0.458	0.447	0	5.09
<i>ln(assets)</i>	14416	21.765	21.641	1.192	16.161	27.386
<i>ln(employees)</i>	14406	7.567	7.506	1.193	1.099	12.302
<i>ln(sales)</i>	14401	21.092	20.977	1.421	11.294	27.621
<i>admin_ratio</i>	13961	0.119	0.105	0.074	0.003	0.912
<i>admin_int</i>	14109	0.103	0.09	0.066	0.013	0.553
<i>kl</i>	14398	12.486	12.491	0.959	3.806	17.897
<i>skill_int</i>	13066	0.22	0.174	0.161	0.002	0.946
<i>intang_ratio</i>	14090	0.187	0.158	0.13	0.002	0.728
<i>CEO tenure</i>	14370	5.132	4.000	3.469	0.000	21.000
<i>age</i>	14417	2.674	2.708	0.417	0	3.932
<i>leverage</i>	14416	0.4	0.381	0.214	0.047	0.985
<i>duality</i>	14281	0.305	0	0.46	0	1
<i>largestshare</i>	14417	0.344	0.324	0.145	0.024	0.99
<i>soe</i>	14417	0.032	0	0.111	0	0.875
<i>investment/assets</i>	12912	0.055	0.032	0.072	-0.132	0.493
<i>cash flow/assets</i>	13113	0.082	0.073	0.06	-0.079	0.374
<i>long_loan</i>	11513	398.022	16.518	1215.098	0	8692.901
<i>short_loan</i>	11513	1019.502	286.6	2284.569	0	15130
<i>guarant_loan</i>	8248	698.117	190	1427.342	0.008	8826.965
<i>credit_loan</i>	7377	1043.813	172	2935.173	0.001	20885.33
<i>mortgage_loan</i>	6443	233.756	86.1	463.015	0.02	3177.312

Note: Units of bank loans are in a million yuan. The definitions of all variables are explained in the data appendix

⁸Unfortunately, we do not have enough observations for pledge loans to conduct empirical analysis.

4 Empirical Strategy

4.1 Selection Bias and the Methodology

Admittedly, as a business strategy, the choice of DT is endogenous.⁹ Simple comparisons of performance between digitalized firms and non-digitalized firms are problematic because intrinsically better firms may have adopted a particular digitalization strategy, thereby making it difficult to conclude that the performance difference is caused by the choice of DT. If some unobserved factors determining the selection into digitalization are not properly controlled, we can not identify the impact of digitalization on firm performance from their ex-ante differences. This is called the selection bias problem.

We tackle the problem of selection bias by employing the panel data structure and rich firm-level characteristics. First, thanks to the panel data of firms, we observe firm performance before and after its digitalization, therefore we can use the differenced outcome instead of outcome levels. Then we compare the difference in the differenced outcomes between digitalized and non-digitalized firms to recover the causal impact of digitalization on firm performance. This is an attractive estimator because it relies on weaker versions of conditional independence assumption, which permits selection to be based on potential outcomes and allows for selection on unobserveables (Heckman et al., 1997). Second, we employ rich information in the data to control for firm-level characteristics that may influence firm's digitalization strategy.

In our data, firms initiate DT in different periods (see Table 1). This staggering manner of treatment adoption motivates us to use econometric techniques developed for staggered DID design. The central challenge thus becomes how to construct reliable comparison groups between digitalized units and non-digitalized units. We tackle this problem by employing the recently developed Diff-in-Diff estimator developed for the staggered DID design. In the appendix, we provide a detailed discussion on the staggered DID design and the Callaway-Sant'Anna estimator (Callaway and Sant'Anna, 2021) that we employ in our analysis.

4.2 Empirical Models

We use the Callaway-Sant'Anna staggered DID estimator to test Hypotheses H1a-H1b, H2, and H3a-H3d. The choice of dependent variable Y_{it} and moderators are summarized in Table 3 and explained in the following.

Hypotheses H1a and H1b are related to the productivity effects of DT. Regarding Hypotheses H1a and H1b, for the main dependent variables, we have explored a series of firm performance measures including TFP , labor productivity ($\frac{Sales}{Employees}$), and capital efficiency ($\frac{Sales}{Fixed Assets}$). To better understand the impact of DT on firms' inputs and outputs, we also employ total assets ($\ln(assets)$) and total employees ($\ln(employees)$) to measure factor input, and total sales ($\ln(sales)$) to measure the output. Given the fundamental role of Hypotheses H1a and H1b, we have also

⁹See related discussions on the endogeneity of the choice of operating mode by Shaver (1998) and Chang et al. (2013).

considered the potential endogeneity issue of DT. To address the endogeneity concern caused by idiosyncratic component in the unobserved factors, we employ an instrumental strategy by using the share of digitalized rivaling firms in the same industry as an instrument for firm's DT. These results are reported in the robustness check section.

To test Hypothesis H2, we use the capital intensity (*kl*), skill intensity (*skill_int*), and intangibility (*intang_ratio*) to measure production's organizational structure, and administration ratio (*admin_ratio*) and administration intensity (*admin_int*) to measure management's organizational structure.

For the tests on Hypotheses H3a-H3d, we use the CEO tenure (*CEO Tenure*), total assets (*ln(assets)*), firm's age (*age*), and skill intensity (*skill_int*) to test the moderating effects of these variables on the relationship between DT and firm performance. Specifically, we group firms into different cohorts based on their age, CEO tenure, skill intensity, and total assets, and estimate the treatment effects of DT for each cohort.

To test Hypothesis H4, we use long-term loans (*long_loan*), short-term loans (*short_loan*), guaranteed loans (*guarant_loan*), credit loans (*credit_loan*), and mortgage loans (*mortgage_loan*) as dependent variables to measure the impact of DT on firm's external financing. As for further tests on the signaling mechanism, following the literature on financial constraints (e.g., Erel et al. (2015)), we include the interaction terms between *DT* and the cash flow (*cash_flow/assets*) in the standard investment model.

To apply the Callaway Sant'Anna staggered DID estimator, we also need let X_i be a vector of variables capturing firm characteristics that can affect the firm's decision on DT. The key of choosing X_i is to ensure that conditional on the firm characteristics X_i , the parallel trend assumption (Assumption 1) is satisfied. The selection of control variables (X_i) is theoretically and empirically justified to support the identification assumptions of the Callaway Sant'Anna staggered DID estimator. First, we control for ex-ante productivity since firms with different initial productivity levels may have varying propensities to adopt DT, ensuring that observed effects are attributable to the DT itself rather than pre-existing productivity differences. Second, we include fundamental firm characteristics: age captures organizational maturity and accumulated capabilities that could affect both DT decisions and implementation abilities; total assets control for resource availability and scale effects; and leverage ratios account for financial constraints that might influence both the capacity to invest in DT and productivity outcomes. Third, our corporate governance variables (duality, largest shareholding, and SOE status) are particularly relevant as they influence decision-making processes, affect the speed and nature of organizational change, and capture potential agency issues that might impact both the decision to digitally transform and its implementation effectiveness. Finally, we include industry dummies because different sectors face varying competitive pressures to digitalize, the potential for productivity gains from DT varies across industries, and industry-specific trends might confound the treatment effect. This comprehensive set of controls strengthens the parallel trends assumption by accounting for observable characteristics that could create selection bias in DT

adoption, with our pre-treatment trend tests providing empirical validation of this approach.

Table 3: Hypotheses and Dependent Variables in Empirical Models

Hypotheses	Dependent Variables	Moderators
H1a, H1b	$TFP, \frac{Sales}{Employees} \text{ and } \frac{Sales}{Fixed Assets}$	n.a.
H2	Production: $kl, skill_int$, and $intang_ratio$ Management: $admin_ratio$, and $admin_int$	n.a.
H3a	TFP	$CEO Tenure$
H3b	TFP	$\ln(assets)$
H3c	TFP	age
H3d	TFP	$skill_int$
H4	$long_loan, short_loan, guarant_loan,$ $credit_loan, mortgage_loan \text{ and } invest-$ $ment/assets$	$cash\ flow/assets$

Note: The hypotheses are explained in Section 2, and the definitions of all variables are described in Section 3.2 and listed in the data appendix. For all empirical models, the main explanatory variable is the DT variable DT , and control variables are $age, total\ assets, leverage\ ratio, duality, largestshare$, and soe .

Based on the group-time specific treatment effects estimator (A.4), we estimate the event-study-type causal parameter $\beta(h)$ using (A.2) for non-negative values of h , and the aggregate effect using (A.3). To rigorously evaluate the parallel trend assumption, we estimate $\beta(h)$ for negative values of h , where insignificant estimates indicate the satisfaction of this crucial assumption. We carefully select a symmetric time window of $-6 \leq h \leq 6$ for several methodological considerations. First, this window provides sufficient pre-treatment periods to establish the validity of the parallel trends assumption while allowing enough post-treatment periods to capture both immediate effects and medium-term dynamics of DT. Second, this balanced window helps mitigate potential bias from selective timing of treatment adoption while ensuring adequate sample size in both early and late periods. Third, it addresses the trade-off between estimation precision and treatment effect identification: a wider window would reduce the number of firms with complete data throughout the period, while a narrower window might miss important dynamic effects of DT. Following Sant’Anna and Zhao (2020), we implement the improved doubly robust DID estimator to enhance the efficiency of our estimates. Furthermore, we employ the multiplier bootstrap procedure proposed by Callaway and Sant’Anna (2021) to conduct asymptotically valid inference, which is particularly crucial given our relatively long time window and the potential for time-varying treatment effects.

We use the STATA program *csdid* to perform the estimation. In our main results, we estimate the event-study-type treatment effects $\beta(h)$ and conduct inferences. We focus on an unbalanced panel with firms that are treated no earlier than the year 2011 and no later than 2018, and firms that are never treated. When using panel data, the estimator does not require data to be strongly balanced. However, note that when estimating each $\beta(e, t)$, only observations that are balanced within a specific 2×2 design are used for the estimator.

As elucidated by Marcus and Sant’Anna (2021) and Borusyak et al. (2021), when deciding

on parallel trends assumption, we face the classical “robustness” versus “efficiency” trade-off: With stronger parallel trends assumption that includes not-yet-treated units, practitioners can potentially achieve gains in efficiency, which leads to more precise inferences. Considering this, we also employ the estimator proposed by Callaway and Sant’Anna (2021) with not-yet-treated units as controls. In addition, to gauge the robustness of our results, we implement estimators developed by Sun and Abraham (2021), de Chaisemartin and d’Haultfoeuille (2020), and Borusyak et al. (2021). These estimators rely on alternative parallel trends assumptions on the not-yet-treated units, which are explained in detail in Marcus and Sant’Anna (2021) and Borusyak et al. (2021).

5 Results

5.1 Treatment Effects on Productivity, Inputs, and Outputs

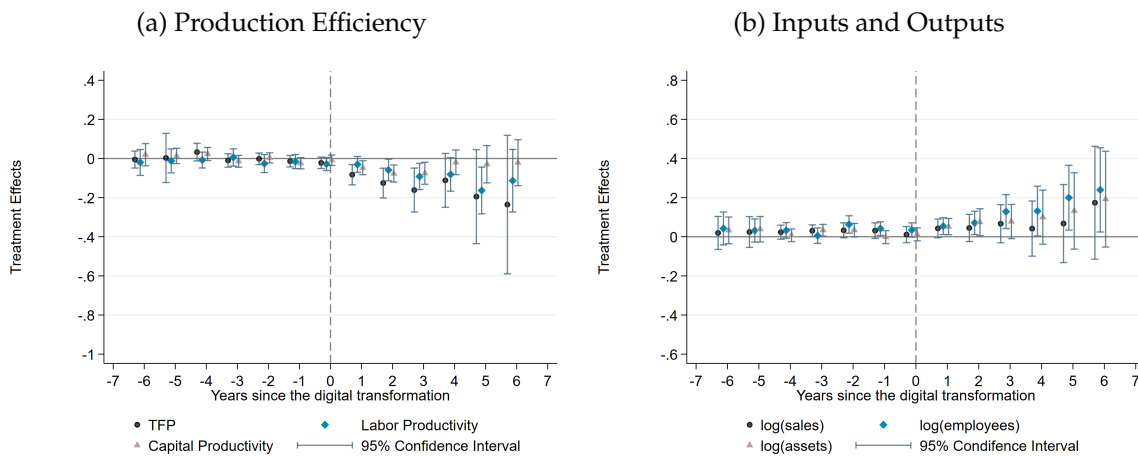
We first test the predictions in Hypotheses H1a and H1b. To avoid the small sample size issue, we report the point estimates with 6 treatment leads and 6 treatment lags and the associated 95% confidence intervals. For each dependent variable. We obtain 12 different estimates using the Callaway and Sant’Anna (2021) estimator, one for each time period h ($-6 \leq h \leq 6$). Figure 1 displays the event-study results for the treatment effects of digitalization on production efficiency, firm inputs, and outputs. The results for production efficiency are reported in *Panel (a)*. We use revenue TFP as the baseline measure for productivity. We see that the coefficient estimates before the treatment are insignificantly from zero, indicating a strong pretreatment parallel trend. In 3 years after digitalization, firms experience a gradual and significant decrease in revenue TFP. The productivity drop is around 10% after one year of digitalization, and approximately 18% after three years. We do not see the revenue TFP to be significantly higher than the pre-digitalization level even after 6 years of digitalization. We also use labor productivity ($\log(\text{Sales}/\text{Total Employees})$) and capital productivity ($\log(\text{Sales}/\text{Total Fixed Assets})$) as alternative measures of firms’ production efficiency. The estimation results are highly consistent, suggesting that our results are robust to different measures of firm productivity. Overall, these results provide strong support for Hypothesis H1a, indicating that DT has led to a significant temporary decline in firm productivity. For Hypothesis 1b, within a window of 6 years after DT, we do observe a significant recovery in productivity after the initial decline. This suggests that firms may need a substantial adjustment period to adapt to new digital technologies and organizational routines, which is largely consistent with Hypothesis H1b.

As productivity can be understood as the ratio between outputs and inputs that reflects the scale of firm’s operation activities, we continue to examine the impacts of DT on firm’s total sales, total assets, and total employees. *Panel (b)* in Figure 1 shows the estimation results. We notice that the parallel trend assumption is well satisfied. We see that the treatment effects of digitalization on sales is not significantly different from zero. This implies that firms may not harness the new digital technologies well enough to boost its sales. In contrast, we see a steady

increase in total employees and total assets. One year after the DT, the digitalized firms increase their total employees and total assets by around 5 percentage points. The growth trend in assets and employees is steady and carries over time. This pattern suggests that DT necessitates substantial and ongoing investments in human capital and physical assets as firms learn to adapt to new technologies. Based on these findings, the persistent productivity decline, rather than being a temporary adjustment phenomenon, appears to stem from this sustained expansion of inputs without corresponding growth in outputs, indicating that firms face significant challenges in effectively leveraging their digital investments to enhance operational efficiency. These findings contradict the notion of a short-term adjustment period and instead point to more fundamental challenges in translating digital investments into productivity gains.

The pronounced three-year decline in productivity followed by a gradual stabilization can be attributed to several key factors in the DT process. Initially, as firms implement digital technologies, they face a steep learning curve where substantial resources are allocated to adaptation-hiring new workers and acquiring assets-without immediate returns in output. This learning phase is particularly intensive in the first three years, as evidenced by the sharp 18% drop in revenue TFP, during which firms must simultaneously manage new digital systems, train existing employees, integrate new specialized personnel, and reconfigure organizational processes. The stabilization of productivity after year three, while not returning to pre-digitalization levels, suggests that firms begin to establish new operational routines and develop digital capabilities. This timing aligns with typical organizational learning cycles, where the most severe disruptions occur during the initial implementation and learning phase. However, the persistence of lower productivity levels even after six years, combined with sustained higher levels of employment and assets, indicates that DT fundamentally alters firms' production functions toward a more resource-intensive operational model, rather than delivering the efficiency gains often anticipated from technological advancement.

Figure 1: Event-study Estimators of Treatment Effects of Digitalization



Note: All standard errors are obtained using the multiplicative bootstrap procedure with 1,000 bootstrap draws.

Table 4 shows the aggregate treatment effects on digitalization on productivity, total inputs, and total outputs. It includes control variables categorized into firm characteristics and financial variables in calculating the propensity to digitalize. Firm characteristics consist of the firm's age, the leverage ratio, and governance indicators such as a binary variable for CEO duality, the shareholding percentage of the largest shareholder, and a binary variable for state-owned enterprises. The Logit model for estimating the propensity score is reported in Table B.5. Table 4 Panel (a) reports the treatment effects on productivity. On average, DT leads *TFP* to decrease by 10.1%, labor productivity by 5.7%, and capital productivity 4.1%. All these estimates are significant at 1% significant level, strongly supporting that digitalization has led to declines in productivity. Table 4 Panel (b) displays the estimates for total inputs and outputs. DT significantly increase total assets by 5.9%, and total employees by 7.9%. In contrast, there is no strong evidence supporting that digitalization has led to growth in total sales. This input-output decomposition supports the interpretation that the productivity decline is driven by the increase in inputs without a corresponding increase in outputs. As a comparison, we also report the TWFE estimates in the Appendix B.2.5.

Table 4: Aggregate Treatment Effects of Digital Transformation

<i>(a) Productivity:</i>			
Outcomes	TFP	$\frac{\text{Sales}}{\text{Employees}}$	$\frac{\text{Sales}}{\text{Fixed Assets}}$
β^{agg}	-0.101*** (0.034)	-0.057*** (0.022)	-0.041*** (0.016)
Controls	Yes	Yes	Yes
Obs.	12412	13503	12420
<i>(b) Inputs and Outputs:</i>			
Outcomes	Assets	Employment	Sales
β^{agg}	0.059** (0.028)	0.079*** (0.025)	0.04 (0.029)
Controls	Yes	Yes	Yes
Obs.	13509	13510	13526

Note: All models include the firm's size, age, corporate governance variables (duality, largest shareholding, and SOE status), and industry dummies. The standard errors are obtained using Wild bootstrap with 1,000 draws and clustered at the firm level. ** $p < 5\%$, *** $p < 1\%$.

Our estimates on the negative impact of DT on productivity is largely consistent with the anecdotal evidence. It is well known that although being massive in its scale, many DTs undertaken by Chinese manufacturing firms fail to achieve their goals. Recently, a joint research by Accenture and CICS-CERT find that only around 11% of Chinese enterprises can be qualified as "Digital Transformation Champions".¹⁰

The baseline estimates consistently show that digitalization has lead to declines in productivity. However, it is worth noting that the productivity drop is a relative change rather than

¹⁰See the full report at <https://www.accenture.com/cn-en/insights/consulting/china-digital-maturity-index-report>.

an absolute decrease. In other words, our estimation results do not imply that digitalized firms have absolute productivity declines during the sample period. To see this, we plot the trends of three types of productivity in Figure 2. *Panel (a)* shows that, similar to untreated firms, treated firms display a growing trend in total factor productivity during the sample period. This pattern also exists when we consider labor productivity (*Panel (b)*) and capital productivity (*Panel (c)*).

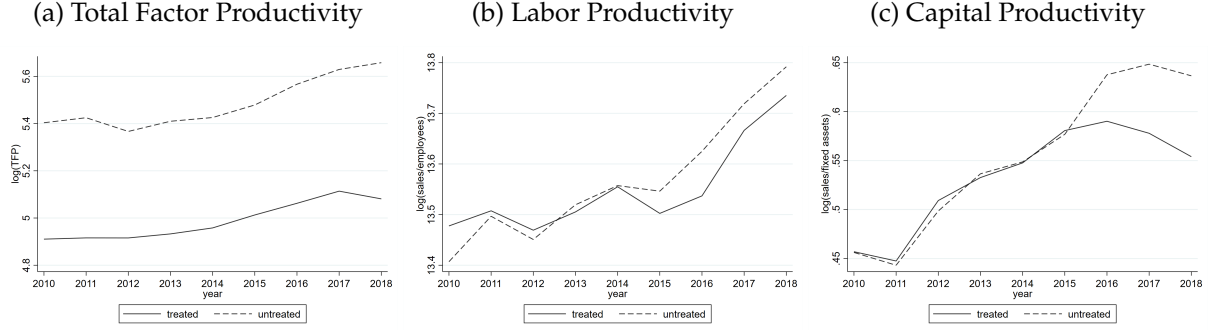
5.2 Robustness

Before we perform richer analysis on the heterogeneous treatment effects and underlying mechanisms, we conduct a series of robustness checks to validate our baseline empirical findings.

5.2.1 Trend Explanation

The negative productivity effects of DT does not necessarily imply that digitalized firms have productivity decline over time. Figure 2 presents the simple trend of productivity for the treated and untreated firms. Overall, we see strong growing trend for these three different measurements of production efficiency. As shown in *Panel (a)*, the total factor productivity of digitalized firms has been increasing over time, similar to the untreated firms. This pattern is also observed in labor productivity (*Panel (b)*). The trend of capital productivity is more volatile, but the treated firms also show a general growing trend in capital productivity before the year of 2016 (*Panel (c)*). However, the productivity declines for both the treated and untreated firms after 2016. The divergent patterns across productivity measures warrant careful interpretation. While TFP and labor productivity show evidence of adaptation and learning effects, the distinct drop in capital productivity after 2016 can be attributed to two main factors. First, the gradual accumulation of digital assets over time - as firms continuously invest in new technologies and infrastructure - leads to a larger denominator in the capital productivity ratio (sales/total fixed assets). This buildup of assets, combined with the time lag required for these investments to generate corresponding sales growth, naturally puts downward pressure on capital productivity. Second, this effect was likely exacerbated by the industrial policy stimulus that increased capital investment in manufacturing sector in China.

Figure 2: Trends of Productivity



Note: labor productivity is measured as the ratio of total sales to total number of employees, and capital productivity is measured as the total sales to total fixed assets.

5.2.2 Placebo Tests

To alleviate the concern of finding false-positive results, we perform two placebo tests. In the first placebo tests, we follow Pierce et al. (2015) and Tan and Netessine (2020) and randomly assign the year of implementing DT for the treated firms, and re-estimate the average treatment effects 100 times. Second, we move the implementing year of digitalization to be several years earlier than the actual treatment time and analyze the dynamic effects of digitalization.

Randomly assigned treatments In Figure 3, we show the estimation results of randomly assigned treatments. Each point represents the average treatment effects defined in (A.2), with the capped spikes being their 95% confidence intervals. We rank the 100 placebo point estimates together with the solid blue line indicating the baseline estimate. For the treatment effects on TFP (Figure 3 (a)), almost all the 95% confidence intervals cover zero and point estimates are well above the baseline estimates. For total assets (Figure 3 (b)), nearly all the 95% confidence intervals of the point estimates include zero and only one point estimate is above the baseline estimates. The simulation results for total employees Figure 3 (c) show that all point estimates are below the baseline estimate and zero lies in almost all confidence intervals. These results for the placebo tests confirm the baseline finding that digitalization decreases productivity, and increase total assets and total employees. Figure 3 (d) displays the estimates using logged total sales as the dependent variable. Around 20 estimates have values close to the baseline results, and almost 20% of the 95% confidence intervals for the estimates include baseline estimates. Recall that we do not find significant treatment effects on total sales. Therefore, the simulation results indicate that the impact of digitalization is no stronger than randomly assigned treatment.

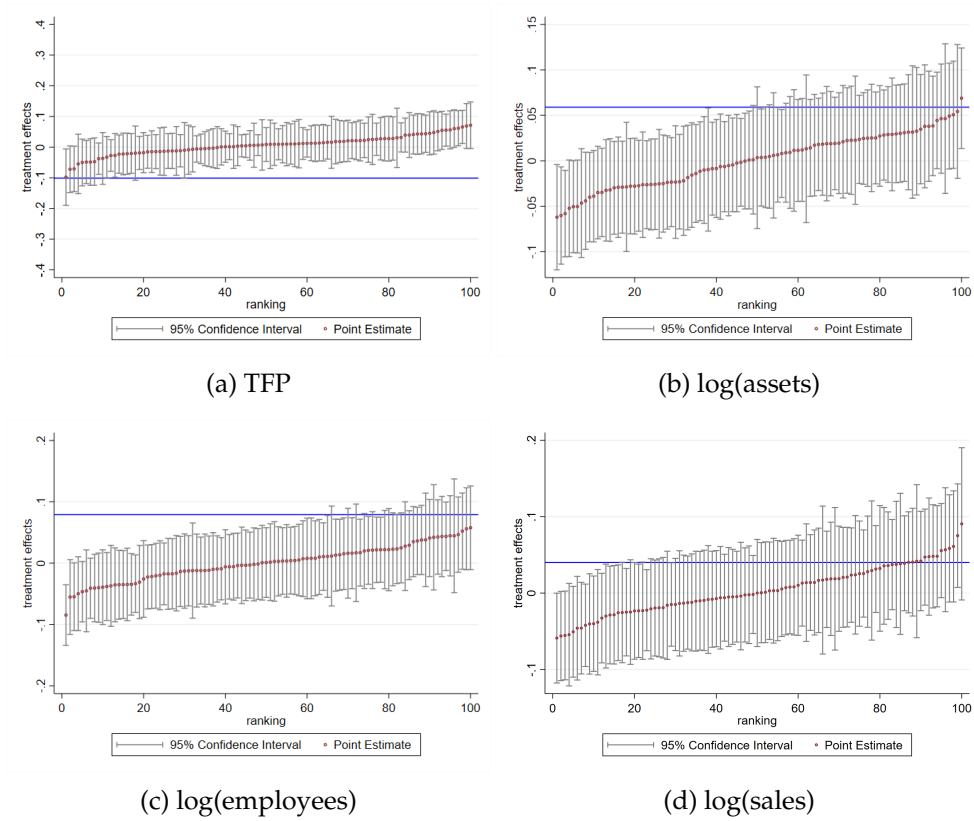
Anticipation Effects In our empirical strategy, we assume that firms have no anticipatory response in productivity to the DT implemented in the future. This assumption can be violated if the CEO and managers foresee the upcoming challenge in DT and undertake some measures

to prepare for it. To circumvent the anticipatory effects, we redefine the group variable to be several years ahead of the period when the DT is first announced in the annual report. In particular, we define $\tilde{T}_i(e)$ to be a dummy variable that is equal to one if firm i first announces the DT in period $e + a$. We choose a to be 1, 2, and 3, and we estimate the group-year specific treatment effects:

$$\tilde{\beta}(e, t) = \mathbf{E}(Y_{it} - Y_{it}(0) | \tilde{T}_i(e) = 1).$$

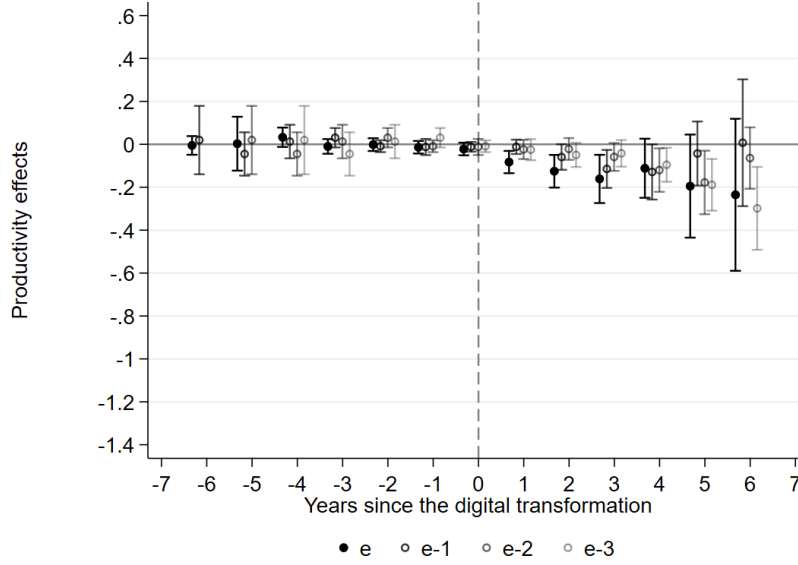
Then the horizon-specific treatment effects are obtained using the aggregation formula (A.2). If firms have productivity responses in anticipating the DT three years earlier, the estimated causal effects should be significant.

Figure 3: Placebo tests



Note: Points indicate the point estimate and capped vertical lines are the 95% confidence interval.

Figure 4: Event-study analysis of TFP: Anticipatory Effects



Note: The dots indicate the point estimates and the capped vertical line displays the 95% confidence interval. All standard errors are clustered at the city level.

Figure 4 shows the event-study analysis of the redefined treatment variable. As we redefine treatment time to be 1 to 3 years earlier than the observed DT, we find no significant productivity effects in the first year after digitalization. Specifically, the significant productivity impacts only emerge after the actual announced digitalization takes place. These results rule out the possibility that productivity has anticipatory responses to digitalization.

5.2.3 Productivity Spillovers of Digital Transformation

Presumably, a firm's performance is not only affected by the firm's own digitalization strategy, but also can be influenced by its rival firms and upstream and downstream firms. DT can generate spillover effects through strategic interactions and supply chain spillovers. Take AI for example, machine learning methods allows firms to make more precise prediction on future market demand (Gans, 2022a,b) and set price flexibly. For non AI-adopting firms, though they can use AI technologies to set prices, they may infer market conditions from its competitors who have adopted machine learning algorithms. Considering this, the "never digitalized" firms are affected by other firms' DT. In a similar spirit, the "not-yet digitalized" firms can also receive spillovers from the already digitalized firms. Although these spillover effects of DT are not explicitly analyzed in our estimation, we expect that they will not overturn our main findings for the following reason. As shown by Butts (2021), in the presence of spillover effects, the estimated treatment effects can be decomposed into (A) direct effect on the treated units, (B) spillover on treated units, and (C) spillover on control units. In particular, our estimated can be thought as (A)+(B)-(C). As illustrated by the theory of complementary and strategic fit,

digitalized firms may do better at assimilating the spillovers from other firms' action of DT than control units who have not started digitalization yet. If the spillover effects are positive, part (B) will be greater than part (C), hence the net spillover effect is positive. This means that our treatment effects estimates for productivity will be more negative if we consider spillovers. On the other hand, if the spillover effects are negative, (B) should be large enough so that the estimated treatment effect is negative despite that (A) is positive. However, at average level, this still implies that digitalization can cause overall negative productivity effects. This only alters the explanation of the empirical result to be re-centering on the negative spillover effects of digitalization.

Since the long-term effects disappear, it is highly likely that the continuous increase in part (C) has offset the positive impacts of parts (A) and (B). However, the current data lacks the granularity and scope needed to rigorously test this hypothesis. Empirically examining the three parts of spillover effects-namely, parts (A), (B), and (C)-is indeed challenging due to the complexity of isolating their individual impacts. We leave this as a potential avenue for future research to explore the spillover effects of DT on firm productivity.

5.2.4 Other Robustness Checks

In Appendix B.3, we perform a series of robustness checks. First, we account for more detailed information on digital transformation, including continuity, content, and intensity; Second, we perform a robustness check related to the timing of treatment. Third, we consider alternative control-group constructions, and clustering levels; Fourth, we use alternative staggered-DID estimators.

5.3 Mechanism Analysis

5.3.1 Impact of DT on Organizational Structures

We first test the prediction in Hypothesis H2. *Panel (a)* in Table 5 shows the results for the treatment effects estimates on the firm's management-related organizational structure. There is no evidence for any decreases in administration works, whether it is measured by the share of administrative employees to total employees or administrative expenditures over total sales.

Panel (b) in Table 5 displays estimates for various measures of the production's organization structures. Again, we see no evidence supporting that digitalization has led to significant changes in the firm's production technology. First, the capital-labor ratio ($PP\&E/Total\ Employees$) is not significantly affected by the digitalization, indicating the DT has not led to biased technical change towards more capital-intensive technologies. Second, the firm's skill intensity ($Skilled\ Workers/Total\ Employees$) is not significantly influenced by DT. As the skill intensity speaks to the firm's skill components of production, the estimation result suggest that firms has not changed the skill requirement after DT. Lastly, considering that during DT firms need to purchase new pieces of software, we also examine the impact of digitalization on the firm's

intangibility ($Intangibles/(Intangibles+PP\&E)$). Still, we find no supporting evidence that digitalization has lead to changes in the firm’s intangibility. Overall, these results show that digitalization does not incur profound adjustments in the firm’s organizational structures. Therefore, our finding of negative productivity impacts is not likely to be driven by temporary adjustments in organizations, which lead to productivity losses. These results are highly consistent with the prediction of Hypothesis H2.

Table 5: Impact of Digitalization on the Organizational Structure

Outcomes	(a) Management		(b) Production		
	$\frac{Admin. Workers}{Total Employees}$	$\frac{Admin. Exp.}{Total Sales}$	$\frac{PP\&E}{Total Employees}$	$\frac{Skilled Workers}{Total Employees}$	$\frac{Intangibles}{Intangibles+PP\&E}$
β^{agg}	-0.005 (0.004)	0.002 (0.002)	-0.004 (0.027)	0.002 (0.005)	0.012 (0.006)
Controls	Yes	Yes	Yes	Yes	Yes
Obs.	12004	12277	12412	11215	12166

Note: The standard errors are obtained using Wild bootstrap with 1,000 draws and clustered at the firm level.

5.3.2 Firm-level Moderators

To uncover the possible mechanisms through which DT influences firms’ productivity, we group firms based on the firm-level moderators as discussed in Hypotheses H3a-H3d estimate the treatment effects separately. Table 6 reports the estimation results based on a series of firm-level factors. First, we consider the tenure of CEO (*Panel (a)*, Table 6) in moderating the impacts of DT on productivity. The median value of CEO’s tenure for treated firms is 4 years, we thus define a CEO has a short-tenure if it is shorter than 4 years.¹¹ Interestingly, we find negative productivity impacts are mainly borne by firms managed by short-tenure CEOs. This implies that more experienced CEOs can deal with the departmental conflicts and challenges incurred by DT better than the non-experienced CEOs, avoiding productivity losses from DT, which supports Hypothesis H3a. We also find that smaller and younger firms tend to have a larger productivity decrease after DT, which are consistent with Hypothesis H3a and H3b, respectively. Lastly, we find that high-skilled firms tend to have a larger decline in productivity after DT. This reflects that employees in high-skilled firms may have a strong resistance to new digital technologies, which supports Hypothesis H3d. Overall, these results are consistent with our theoretical predictions based on the 3-pillar framework proposed by Scott (2001).

¹¹For size, age, and skill intensity, we use a similar approach to group firms.

Table 6: Firm Characteristics and Treatment Effects on Productivity

	(a) CEO Tenure		(b) Size	
	Short	Long	Small	Big
β^{agg}	-0.157*** (0.058)	-0.073* (0.040)	-0.153** (0.072)	-0.112** (0.046)
Obs.	10792	10941	10835	10640
	(c) Age		(d) Skill Intensity	
	Young	Old	Low	High
β^{agg}	-0.133** (0.057)	-0.092** (0.043)	-0.072** (0.030)	-0.158** (0.064)
Obs.	10919	10814	10849	10884

Note: The standard errors are obtained using Wild bootstrap with 1,000 draws and clustered at the firm level. ** $p < 5\%$, *** $p < 1\%$.

5.3.3 Financial Motives for Digitalization

We have confirmed that digitalization leads to negative impacts on productivity and barely affects firms' organizational structures. These seemingly negative results trigger us to further think the motives for digitalization strategy. As an emerging economy, China is characterized by capital market imperfections. Although it has been growing, the Chinese stock market is relatively small compared with the banking sector (Guariglia and Yang, 2016). Especially for State-Owned enterprises, bank loans remain as an important source of external credits. In sharp contrast, non-state-owned enterprises have limited access to banking credit and rely more on internal financing (Liu et al., 2021). In the sample, the average proportion of state-owned shares is 0.038, with the 75th percentile being 0 and the 90th percentile being 0.094, which indicates that most of the digitalized firms are non-state-owned. Under this environment, can DT be employed as a development strategy to gain external financial support? Considering this, we further investigate how digitalization affects the firm's borrowing from banks.

Usually, the interest rate on short-term loans is lower than that on the long-term loans because the lender takes on more risk of defaulting for long-term loans. Therefore, it is usually more difficult for firms to receive long-term bank loans. Table 7 shows that DT leads firms to gain more long-term banking credit (around a 191.550 million yuan increase). However, we do not find a significant increase in short-term bank loans. These results reflect that digitalization gives firms advantages in obtaining banking credits.

We also examine the treatment effects of DT on loans using different types of collaterals. Specifically, we employ credit loans, guaranteed loans, and mortgage loans as dependent variables.¹² Both guaranteed loans and mortgage loans are secured loans, and are of lower risks

¹²A *credit loan* is a loan granted on the basis of the borrower's own creditworthiness. It does not require the debtor to provide collateral or third-party guarantee, and the borrower's credit level is used as the repayment guarantee. *Guaranteed loans* refer to the loans granted based on a third party's commitment. *Mortgage loans* require the borrower to provide a certain amount of collateral as security to ensure that the loan is repaid at maturity. Collateral is generally easy to preserve, not easy to wear, easy to sell goods, such as securities, notes, stocks, real estate, etc. When the loan expires, if the borrower fails to repay the loan on time, the bank has the right to auction

than credit loans. However, firms bear less risk when borrowing guaranteed loans because the guarantor will be the final payers in case of defaulting. In Table 7, we find significant increases in credit loans and guaranteed loans, but not mortgage loans. These results confirm our conjecture that banks and guarantors treat digitalization as a good signal for potential growth.

Table 7: Digitalization and Bank Loans

Types	Total	(a) Loan Terms		(b) Collateral Types		
		Long-Term	Short-Term	Credit	Guarantee	Mortgage
β^{agg}	367.581** (143.882)	191.586*** (61.096)	125.101 (84.450)	281.827** (152.702)	201.730*** (78.577)	61.908 (40.102)
Obs.	10552	10552	10552	6421	7381	5595

Note: bank loans are in millions of RMBs. The standard errors are obtained using Wild bootstrap with 1,000 draws and clustered at the firm level. ** $p < 5\%$, *** $p < 1\%$.

One may infer that digitalized firms may obtain more bank loans to finance its investments in digital technologies. In this case, the firm's financial constraints should not be relaxed after digitalization. In addition, the borrowing interest rates should be increasing as banks can seize this opportunity to extra more rents from firms. On the other hand, if firms use digitalization to send signals indicating growth potential, this may help relieve the information asymmetry problem between banks and the firm, which in turn ease the firm's financial constraints through issuing more bank loans. We conduct two tests to (in)validate these two competing explanations. First, we examine the impact of digitalization on firm's financial constraints. Second, we investigate the change in borrowing interest rate after digitalization.

In a large body of literature starting with Fazzari et al. (1988), the estimates of investment-to-cash flow sensitivity is used to measure financial constraints. The idea is that, in the absence of financial constraints, a firm should take all value-enhancing investments regardless of the firm's financial condition. However, when the firm is financially constrained, then the projects it can undertake will be rationed. As a result, an increase in the cash flow will lead the firm to undertake more projects. To evaluate the impact of the digitalization on firm's financial constraints, we follow Erel et al. (2015) to estimate the following specification:

$$\frac{Investment_{it}}{Assets_{it-1}} = \beta_0 + \beta_1 \frac{Cash Flow_{it}}{Assets_{it-1}} + \beta_2 Digit_{it} \times \frac{Cash Flow_{it}}{Assets_{it-1}} + \beta_3 Digit_{it} + \mathbf{Z}'_{it} \gamma + \lambda_i + \lambda_{pt} + u_{it}, \quad (2)$$

where $Assets_{it-1}$ is the total assets, $\mathbf{Z}'_{it}$ is a vector of control variables, λ_i is the firm-level fixed effects, λ_{pt} is the province-year fixed effects, and u_{it} is the error term. In Equation (2), the sensitivity of gross investment to cash flow is given by $\beta_1 + \beta_2 Digit_{it}$. We are interested in the value of β_2 . A negative β_2 indicates that the digitalization relieves firms' financial constraints. Table 8 shows the estimation results. We find that the coefficient of the interaction the collateral and repay the loan with the proceeds from the auction.

term ($Digit_{it} \times \frac{Cash\ Flow_{it}}{Assets_{it-1}}$) is negative and statistically significant at 5% in column (1), and remain significant at 10% in column (2). These estimates suggest that digitalization acts as a signal to banks lenders who help ease the firm's financial constraints.

We also examine the impact of digitalization on the borrowing interest rate. Because we do not observe the interest rate directly, we use the interest payable to total bank loans as a proxy. Our estimate show that the average treatment effect is $-.001$ with a standard error of $.002$. The insignificant treatment effect shows that the interest rate facing firms are not increasing despite that the firms increase the amount of riskier bank loans (long-term loans and credit loans). Therefore, it is not likely that the increase in bank loans is mainly driven by the investments demand brought up by digitalization. Instead, these findings suggest that firms can strategically signal its growth potential through digitalization to earn banks' financial supports.

Table 8: Digitalization and Financial Constraints

Dependent Variable	Investment/Total Assets	
	(1)	(2)
<i>Cash Flow/Total Asset</i>	0.331*** (0.057)	0.249*** (0.059)
<i>Digital</i>	0.007 (0.006)	0.007 (0.006)
<i>Digital</i> $\times$ <i>Cash Flow</i>	-0.115** (0.056)	-0.091* (0.054)
<i>Ln(Total Asset)</i>	0.024*** (0.005)	0.017*** (0.005)
<i>Sales Growth</i>		0.037*** (0.009)
<i>Age</i>		-0.069*** (0.021)
<i>Leverage</i>		0.030 (0.019)
<i>Constant</i>	-0.507*** (0.114)	-0.183 (0.130)
adj. R^2	0.217	0.241
Observations	3005	3004

Note: All firm fixed effects and province-year fixed effects are included in all regressions. The standard errors are clustered at the firm level and associated standard errors are in parentheses. * $p < 1\%$, ** $p < 5\%$, *** $p < 1\%$.

6 Conclusion

6.1 Discussion of Main Findings

Our study examines how DT affects the productivity of Chinese manufacturing firms from 2010 to 2018 and yields three main findings. First, DT can, on average, reduce firm productivity by

approximately 10%. More importantly, our evidence does not support positive long-term productivity effects of DT within the observed timeframe. We do not observe a significant productivity rebound during the sample period, nor do we find evidence that formal organizational-structure changes mediate a transition from negative to positive effects. Taken together, these results suggest that the productivity consequences of DT are not limited to initial implementation costs. Rather, they appear to reflect more persistent coordination frictions, incomplete capability conversion, and difficulties in aligning digital initiatives with core operating routines.

Second, the productivity consequences of DT vary systematically with firms' internal conditions. Drawing on Scott's three institutional pillars, we find that CEO tenure, firm size, firm age, and skill intensity shape firms' ability to manage digital disruption. Firms led by short-tenured CEOs, smaller firms, younger firms, and firms with more skill-intensive workforces experience larger productivity losses following DT adoption. These patterns suggest that executive socialization, governance capacity, organizational experience, and knowledge structure all matter for whether digital investments can be translated into productive outcomes.

Finally, we identify a financial signaling effect of DT. Even when DT does not improve productivity within the observed period, firms that adopt DT are better able to relax financial constraints and secure long-term bank loans. This suggests that DT functions not only as an operational initiative but also as a strategic signal to external stakeholders. In environments marked by information asymmetries and imperfect capital markets, DT may convey modernization, managerial commitment, and future growth potential, thereby improving firms' access to external finance.

6.2 Theoretical Implications

Our study offers a more integrated perspective on DT by theorizing DT as a multi-stage organizational process in which capability formation, institutional embeddedness, and stakeholder evaluation jointly shape value creation. This perspective extends existing management research in three ways.

First, our study advances research on DT by challenging the logic that underlies both the classic IT-value literature and more recent work on DT: namely, that digital adoption leads to organizational upgrading and, ultimately, superior performance (Brynjolfsson and Hitt, 1996, 2000; Hanelt et al., 2021). In our setting, DT is associated with a persistent productivity decline rather than an observable productivity gain within the sample period. This finding refines dynamic capability theory by showing that DT may initiate capability formation without generating capability payoffs within the observed period. While prior work has used dynamic capabilities to explain how firms sense opportunities, seize them, and reconfigure resources under digital change (Teece, 2018; Warner and Wäger, 2019; Simsek et al., 2024), our results suggest that such reconfiguration is better understood not as an already realized organizational asset, but as a costly, uneven, and uncertain process of capability construction. Our

study shifts the literature from asking whether DT reflects dynamic capabilities to revealing when capability-building efforts become performance-enhancing.

Second, our study contributes to institutional perspectives by showing that the consequences of DT depend systematically on firms' internal institutional conditions. Prior research has recognized the importance of organizational context in digital change (Scott, 2001; Leonardi, 2012; Wessel et al., 2021), but has paid less attention to how internal institutional arrangements shape the performance effects of DT. Our findings show that CEO tenure, firm size, firm age, and skill intensity capture differences in executive socialization, governance capacity, organizational experience, and knowledge structure. By linking these dimensions to Scott's normative, regulative, and culture-cognitive pillars (Scott, 2001), we show that internal institutional embeddedness is central to explaining why similar digital initiatives produce different outcomes across firms. In this way, our study brings institutional theory into closer dialogue with recent work emphasizing that the value of digital-based tools depends on leadership, experience, and organizational context (Kim et al., 2024; Simsek et al., 2024; Wang et al., 2024; Cui et al., 2026).

Third, our study extends the management literature by introducing a financial signaling perspective on DT. Existing research has primarily treated DT as an operational, strategic, or organizational phenomenon (Hanelt et al., 2021; Yoo et al., 2024). Our findings suggest that DT also has an external evaluative dimension. Even when productivity gains do not materialize within the observed timeframe, DT can improve firms' access to long-term finance. This indicates that the value of DT arises through realized operating outcomes, as well as stakeholder interpretation. DT therefore matters not only because it changes how firms operate, but also because it changes how firms are evaluated by external capital providers.

6.3 Practical Implications

From a practical perspective, our findings offer valuable insights for managers and policymakers aiming to harness the benefits of digital technologies. The negative short-term impact of DT on productivity suggests that firms should be cautious in their approach to digitalization, ensuring that they allocate sufficient resources toward employee training and organizational restructuring to mitigate initial disruptions. For firms in emerging economies, such as China, where financial constraints are prevalent, DT can serve as a strategic tool for signaling growth potential to external investors. By demonstrating successful integration of digital technologies, firms can enhance their creditworthiness and access to external financial resources, which is critical for sustaining firm development in competitive markets.

Additionally, our study highlights the importance of aligning managerial incentives with long-term strategic objectives to achieve sustainable productivity gains. Firms should consider adopting incentive structures that promote strategic foresight and the effective management of DT initiatives. Additionally, our findings suggest that firms with adaptable and strategically aligned organizational structures are better positioned to realize the productivity benefits of DT. Therefore, managers should focus on fostering organizational agility and resilience to

navigate the complexities of DT effectively.

6.4 Limitations and Future Research

Our study acknowledges several limitations that present opportunities for future research to build upon. First, due to data constraints, our investigation of DT effects is confined to a six-year period after the initiation of DT. It is plausible that firms may experience productivity gains over a more extended period, as suggested by literature indicating potential long-term benefits of digitalization (Brynjolfsson et al., 2021; Feigenbaum and Gross, 2021). Future studies could benefit from longer observation periods to better understand these dynamics and to capture potential longer-term productivity improvements. Second, our primary variable of interest—an integrated index of a firm’s digitalization strategy—may not fully capture the nuances and heterogeneity inherent in various types of digital technologies. Future research could delve into the differential impacts of distinct digital technologies or strategies, potentially offering more granular insights into how specific digital initiatives contribute to firm performance. Third, the use of productivity as a proxy for firm efficiency presents its own set of limitations. In our study, productivity is derived from revenue data, which inherently encapsulates both production efficiency and pricing strategies. This duality complicates the isolation of DT’s effects on operational efficiency versus market-driven pricing dynamics. Access to separate data on quantities produced and prices charged could enable future researchers to disentangle these components, allowing for a clearer identification of digitalization’s distinct impacts on production efficiency and pricing strategies.

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Appendices

A Introduction to the Staggered DID Estimator

Traditionally, researchers rely on two-way fixed effects estimator (TWFE) to deal with the staggered DID designs. However, recent advancements in the econometric theory question the validity and the robustness of the TWFE estimator in staggered DID designs (See Athey and Imbens (2018), Goodman-Bacon (2021), Baker et al. (2021), Callaway and Sant’Anna (2021), Sun and Abraham (2021), among others.). As shown by de Chaisemartin and d’Haultfoeuille (2020), Goodman-Bacon (2021), Callaway and Sant’Anna (2021) and Borusyak et al. (2021), the treatment effect coefficient estimator in TWFE can be interpreted as a weighted average of treatment effects in the presence of group- and time-specific treatment effects. To make it worse, some of these weights can be negative, which results in misleading conclusions on the actual impacts of digitalization.¹³

Noticing the problems of the conventional TWFE estimators, Sun and Abraham (2021), Callaway and Sant’Anna (2021), de Chaisemartin and d’Haultfoeuille (2020), and Borusyak et al. (2021) have proposed alternative estimators for obtaining interpretable treatment effects parameters. These estimators differ in the invoked parallel trends assumption and how the potential outcomes are imputed (Marcus and Sant’Anna, 2021). In our sample, around 69% of firms do not experience DT in the time window of consideration. The large size of never-treated firms enables us to make a relatively weak parallel trends assumption. We thus employ the Callaway-Sant’Anna estimator which has the weakest parallel trend assumption that requires common pretreatment trends between treated and never treated groups (Callaway and Sant’Anna, 2021). An additional advantage of this estimator is that it can capture rich sources of treatment effect heterogeneity originating from different treatment cohorts. Importantly, this allows us to study the group-specific treatment effect dynamics.

The Callaway-Sant’Anna Estimator Recall that the binary variable DT_{it} to denote the variable of DT; $DT_{it} = 1$ if firm i has reported DT on later than t and zero otherwise. Define $T_i(e)$ to be a dummy variable that is equal to one if firm i is first treated in period e , and C_i be a dummy variable that is equal to one if firm i is never treated. Let Y_{it} be the outcome variable for firm i in year t and denote $Y_{it}(0)$ as the potential outcome of firm i if it is never treated.

The idea of the Callaway and Sant’Anna (2021) estimator is to estimate some pre-defined “cohort-average treatment effects on the treated” (CATT) and then aggregate them across cohorts and periods to obtain the desired treatment effects estimates. For treated observation (i, t) , the interested average causal treatment effects of digitalization is $\beta_{it} = E[Y_{it} - Y_{it}(0) | D_{is} = 1, s \leq t]$. For those firms first treated at time e , the average treatment effects on the treated

¹³The negative weighting problem is more general than it appears. It also applies to the first-difference specification (de Chaisemartin and d’Haultfoeuille, 2020) and the dynamic specification (Sun and Abraham, 2021).

(ATT) at year t is given by

$$\beta(e, t) \equiv \mathbf{E}[Y_{it} - Y_{it}(0) | T_i(e) = 1]. \quad (\text{A.1})$$

This is the so-called cohort-specific treatment effects on the treated (CATT). We focus on the estimation of $\beta(e, t)$ as it is the cornerstone for the parameters of interests; various types of average treatment effects on the treated (ATT) can be obtained by properly aggregating $\beta(e, t)$. Fundamentally, the challenge of identifying $\beta(e, t)$ stems from missing data on potential outcomes $Y_{it}(0)$ for the treated units. This requires the econometrician to use appropriate control groups to impute the counterfactual outcomes.

Our estimation goal is the event-study-type ATT that is defined for different years after firms undertake DT, separately. It is straightforward that the dynamic ATT is simply a weighted average of the CATT. Formally, we define the estimation target $\beta(h)$ as the ATT for firms that have undertaken DT for h years, then we have:

$$\beta(h) = \sum_{e \in \mathbb{T}} w_e \beta(e, e + h), \quad (\text{A.2})$$

where w_e is the appropriate weights. Lastly, the aggregate treatment effects of digitalization on outcome Y is given by the weighted average of ω_h :

$$\beta^{agg} = \sum_{h \geq 0}^6 \omega_h \beta(h). \quad (\text{A.3})$$

By choosing the correct weights ω_h , the aggregate treatment effects estimator avoids the problem of negative weights in the two-way fixed effects regressions. We follow Callaway and Sant'Anna (2021) to choose the weights ω_e and ω_h , which are the sample sizes of treated firms in the corresponding cohort.

The key identification assumption we require is the following conditional parallel trends assumption.¹⁴

Assumption 1. (*Parallel Trend Assumption*) For all $e, t = 2011, \dots, 2018$, such that $t \geq e$,

$$\mathbf{E}[Y_{it}(0) - Y_{it-1}(0) | T_i(e) = 1, \mathbf{X}_i] = \mathbf{E}[Y_{it}(0) - Y_{it-1}(0) | C_i = 1, \mathbf{X}_i].$$

In the context of our empirical study, the parallel trend assumption requires that, conditional on ex-ante firm characteristics $\mathbf{X}_i$, the potential firm performance $Y_{it}(0)$ of digitalized firms and those never-digitalized firms share a common trend after the DT. This parallel trend assumption allows us to use the “never treated” units as a fixed comparison group. This assumption is relatively weak for several reasons. First, it does not restrict pretrends across different groups. Second, it does not presume that “later” treated units can be used as a valid

¹⁴For a more complete discussion on the underlying assumptions, we refer the readers to Callaway and Sant'Anna (2021).

comparison group for “early treated” units. Third, since the parallel trend assumption is made for the differenced outcome, it is also less restrictive than the parallel trend assumption for level the outcomes. To see this, note that imposing $E[Y_{it-1}(0)|T_i(e) = 1, \mathbf{X}_i] = E[Y_{it}(0)|C_i = 1, \mathbf{X}_i]$ immediately imply Assumption 1, but the reverse is not true. An important implication is that the strategic decision of DT can depend on the potential firm performance variable $Y_{it}(0)$ and $Y_{it-1}(0)$ so long as it is independent of the differenced outcome $Y_{it}(0) - Y_{it-1}(0)$. This alleviates the concern for the self-selection problem that firms may undertake DT by comparing between $Y_{it}(1)$ and $Y_{it}(0)$, which may contaminate the estimates for the actual impact of digitalization on firm performance.

It is worth noting that the parallel trend assumption also permits that firm’s digitalization strategy to depend on some unobserved firm-level factors that are persistent over time. These unobserved factors may include (e.g., unobserved cost heterogeneity, resource constraints, or competitive advantages), that affects the firm’s digitalization strategy. In particular, the parallel trend assumption still holds in the following two typical scenarios. First, these unobserved factors have no impact on the potential outcomes. In this case, it is obvious that the parallel trend assumption is robust to the selection on this type of unobserveables. Intuitively, these unknown factors can only influence potential outcomes through shaping the firm’s digitalization strategy, thus not causing problem for identifying the impact of digitalization on firm performance. Second, when the influential factors are time invariant, using the after-before differenced potential outcome would eliminate them and thus not violating the parallel trend assumption.¹⁵ As a corollary, when the unobserved factor which firms select digitalization on is time varying, if it can be decomposed into a time-invariant component and an exogenous idiosyncratic part, the parallel trend assumption also holds.

Define the generalized propensity score as $P_e(\mathbf{X}_i) = \Pr(T_i(e) = 1|\mathbf{X}_i, T_i(e) + C_i = 1)$, which is the probability of being treated conditional on having covariates $\mathbf{X}_i$ and conditional on being a member of group e or the control group. In a spirit similar to Abadie (2005), Callaway and Sant’Anna (2021) show that the group-time treatment effects $\beta(e, t)$ can be non-parametrically identified as:

$$\beta(e, t) = \mathbf{E} \left[\underbrace{\left(T_i(e) \mathbf{E}[T_i(e)]^{-1} - \frac{P_e(\mathbf{X}_i) C_i}{1 - P_e(\mathbf{X}_i)} \mathbf{E} \left[\frac{P_e(\mathbf{X}) C_i}{1 - P_e(\mathbf{X}_i)} \right]^{-1} \right)}_{\text{weights}} \underbrace{(Y_{it} - Y_{it-1})}_{\text{differenced outcome}} \right]. \quad (\text{A.4})$$

The right-hand side of the equation is a weighted average of the differenced firm-level outcome. The first part of the formula is the weights that depend on the generalized propensity score $P_e(\mathbf{X}_i)$. The intuition for the weights is as follows. For treated observations in group e ,

¹⁵Formally, we can write the potential outcome equation as $Y_{it}(0) = v_i + \epsilon_{it}(0)$, where v_i is the firm-level factor that can be affect the firm’s digitalization strategy. Since we do not observe v_i , we cannot sort out the impact of DT from the unobserved heterogeneity in v_i . However, the differenced outcome $Y_{it}(0) - Y_{it-1}(0) = \epsilon_{it}(0) - \epsilon_{it-1}(0)$, which does not contain the unobserved factor v_i , thus ruling out the endogeneity problem caused by unobserved firm-level factors.

taking never-treated observations from the control group, one weights up (*resp.* down) the controlling observations that have characteristics similar to those frequently (*resp.* rarely) found in group e . This re-weighting procedure guarantees that the covariates are balanced between the treated group e and the control group. Different from the propensity score matching method which matches the treated units with some control units that are close in the propensity score, the re-weighting procedure uses all control units by putting more weights to more similar control units.

B Data Appendix

B.1 Definitions of Variables

Table B.1: Definitions of Variables

Variables	Definitions
<i>Digital</i>	Treatment indicator of digitalization: equal to one since the year when any forms of digitalization are detected in the annual reports
<i>TFP</i>	Logged total factor productivity calculated using LP method following Levishon and Petrin (2003) and by each industry
<i>labor Eff.</i>	Labor productivity calculated as the logged sales per capita
<i>capital Eff.</i>	Capital efficiency calculated as the logged sales per unit of fixed asset; Fixed assets is measured using PP&E
<i>lnasset</i>	Log of total assets = log(book value of total assets) Book value of total assets = Fixed assets + Current assets)
<i>employees</i>	Log of total employees
<i>sale</i>	Log of total annual sales
<i>mana_ratio</i>	Ratio of administration workers to total employees
<i>admin_int.</i>	Ratio of administration expenditures total sales
<i>kl</i>	Capital intensity: the ratio of total fixed assets to total employees
<i>skill_int</i>	Skill intensity: the ratio of skilled workers to total employees; Skilled workers are those with bachelor degree and higher
<i>age</i>	Log(current year- establish year)
<i>leverage</i>	Total debt to total assets
<i>duality</i>	Dummy variable equal to one if CEO is also the general manager
<i>largestshare</i>	Shareholding ratio of the largest shareholder
<i>soe</i>	Proportion of state-owned shares in total shares
<i>CEO tenure</i>	Tenure of the current CEO
<i>Cash flow</i>	Cash and cash equivalents
<i>Gross Investment</i>	Fixed assets- Lagged fixed assets + depreciation

Note: firm-level variables are from CSMAR, Wind Economic Database; annual reports are scrapped from Websites of Shenzhen Stock Exchange, Shanghai Stock Exchange, and CNINF.

B.2 Construction of Key Variables

B.2.1 Measuring Digital Transformation

Keywords Searching As digitalization includes various state-of-art digital technologies, we follow existing studies and choose four groups of keywords (with Chinese bopomofo in brackets) in the following Table B.2:

Refining In some 10-Ks, firms may describe the developing of digitalization in its own sector or China's national strategy, which is not related to the firm's own implementation of digitalization. In our construction of the digitalization indicator, we exclude such scenarios by manually identify them and exclude them as the firm's own digitalization. Following Zhai et al. (2022), we hired two research assistants independently to manually read the extracted texts to determine whether the firm actually undertakes DT in each year. We list three examples below:¹⁶

- **Example 1** (Stock ID: 000008, Year: 2018) “Driven by the trend of technological progress, rail transit operation and maintenance equipment has been upgraded from informatization to digitalization characterized by intelligence, data, internet and deep learning. Traditional equipment is upgrading to intelligent equipment; operation and maintenance system is upgrading from single-product intelligence to unmanned maintenance factory. It is the right time for data-oriented development of equipment in rail transit industry.”
- **Example 2** (Stock ID: 300161, Year: 2017): “*Made in China 2025* puts forward the strategic goal of achieving manufacturing power through ‘three steps’. Centering on the top-level design of *Made in China 2025*, relevant supporting policies have been issued successively, and intelligent manufacturing pilot demonstration projects have been accelerated, with obvious demonstration effect. With the further deepening of transformation, China’s manufacturing industry will be further enhanced in digitalization, networking and intelligence.”
- **Example 3** (Stock ID: 000020, Year: 2012) “...domestic and international economic environment is complex, with difficult concerns and positive factors co-existing. On the one hand, the ability of technological innovation is insufficient. *In the new wave of industrial revolution which centers on global digital and intelligent manufacturing, the gap between domestic enterprises and developed countries in Europe and the United States in the field of high-end technology is facing the risk of being widened again, and enterprises will bear the pain of structural adjustment in the process of industrial upgrading ...*”

The quoted paragraph in *Example 1* talks about the digitalization development in its own industry, but not the firm's own digitalization. In *Example 2*, the paragraph is a description of China's national strategy on digitalization. *Example 3* mentions the global environment of digitalization, but not the firm's own strategy.

¹⁶The English texts are translated from Chinese texts in firms' annual reports.

Table B.2: Expanded Categorization of Keywords into Digital Transformation Areas

Category (ABCD)	Keywords
Artificial Intelligence (AI)	Smartness (Zhi Neng), Intelligence (Zhi Hui), Smart Manufacturing (Zhi Neng Zhi Zao), Smart Factory (Zhi Hui Gong Chang), Smart Production (Zhi Neng Zhi Zao), Smart Firms (Zhi Hui Xing Qi Ye), Machine Learning (Ji Qi Xue Xi), Deep Learning (Shen Du Xue Xi), Computer Vision (Ji Suan Ji Shi Jue), Natural Language Processing (NLP, Zi Ran Yu Yan Chu Li), AI-Driven Decision-Making (AI Qu Dong De Jue Ce), Predictive Analytics (Yu Ce Fen Xi), Autonomous Systems (Zi Dong Hua Xi Tong)
Blockchain	Blockchain (Qu Kuai Lian), Decentralized (Qu Zhong Xin Hua), Distributed Ledger Technology (DLT, Fen Bu Shi Zhang Ben Ji Shu), Smart Contracts (Zhi Neng He Tong), Cryptocurrency (Huo Bi), Tokenization (Biao Qian Hua), Consensus Mechanism (Gong Shi Ji Zhi), Public Chain (Gong Lian), Private Chain (Si Lian), Permissioned Blockchain (Xu Ke Lian), Blockchain-Enabled Supply Chain (Qu Kuai Lian Cang Lian), Traceability (Ke Zhui Zong Xing), Anti-Counterfeiting (Fan Wei Zao), Blockchain-Based Payments (Qu Kuai Lian Fu Kuan), Decentralized Finance (DeFi), Cross-Border Payments (Kuai Jie Guo Ji Zhi Fu), Digital Assets (Shu Zi Zi Chan), Blockchain Platform (Qu Kuai Lian Ping Tai), Decentralized Applications (DApps), Blockchain-as-a-Service (BaaS), Digital Identity (Shu Zi Shen Fen), Immutable Records (Bu Ke Shan Chu De Ji Lu), Permissionless Systems (Wu Quan Xian Xi Tong)
Cloud Computing	Cloud Computing (Yun Ji Suan), Industrial Cloud (Gong Ye Yun), Cloud Platform (Yun Ping Tai), Digital Purchasing Platform (Dian Zi Cai Gou Ping Tai), SaaS, IaaS (Infrastructure as a Service), PaaS (Platform as a Service), Edge Computing (Bian Yuan Ji Suan), Hybrid Cloud (Hun He Yun), Virtualization (Xu Ni Hua), Cloud Storage (Yun Chu Cun), Cloud Security (Yun An Quan), Serverless Computing (Wu Fu Wu Qi Ji Suan)
Big Data	Big Data (Da Shu Ju), Internet of Things (Wu Lian Wang or IoT), Industrial Internet (Gong Ye Wu Lian Wang), C2M (Customer-to-Manufacturer), Data Analytics (Shu Ju Fen Xi), Data Mining (Shu Ju Kuang Jue), Business Intelligence (Ye Wu Zhi Hui), Data Visualization (Shu Ju Ke Shi Hua), Real-Time Processing (Shi Shi Chu Li), Predictive Modeling (Yu Ce Mo Xing), Smart Sensors (Zhi Neng Chuan Gan Qi), Data Warehousing (Shu Ju Cang Ku), Data Governance (Shu Ju Guan Li), Machine-Generated Data (Ji Qi Sheng Cheng Shu Ju), Digital Twins (Shu Zi Shuang Sheng)

Examples of Identified Digitalization We display some examples of texts that are identified as digitalization after performing manual text analysis:

- **Production Digitalization:** Any digitalization expressions related to production, manufacturing and equipment or workshop upgrade are classified as production digitalization.
 - **Example 1**¹⁷ (Stock ID: 002085, Year: 2018): “Our company has intensified the transformation and upgrading efforts, established intelligent factories with robots as the core, improved the automation level of manufacturing industry, and improved the core competitiveness. By building a digital platform in the whole field of digital research and development, digital technology and digital manufacturing of Wanfeng, our company optimized and standardized the operating system, realized the product life cycle management, and provided data support for company information construction and intelligent manufacturing of intelligent factory.”
 - **Example 2** (Stock ID: 002920, Year: 2018): “.....The company has built digital intelligent factory in an all-round way and established industry-leading highly automated and information-based production lines. Now digital intelligent factory and intelligent storage system have been put into use successively. The construction project of integrated industrialization of automobile electronics and mobile Internet technology has officially laid the foundation and is under construction.”
- **Management Digitalization:** Any expressions related to management or the deployment of management information systems are classified as management digitalization.
 - **Example 1**¹⁸ (Stock ID: 002907, Year: 2018): “Our company is carrying out the construction of production automation information management system, and plans to adopt MES system for the company’s IPO and convertible debt raising project ‘the fifth phase of new GMP Production base project’, and build intelligent workshop according to the European and American cGMP standards, and strive to realize the informatization, networking, digitalization and intelligence of production management.”
 - **Example 2** (Stock ID: 002570, Year: 2015): “The company is in the stage of systematic development towards digitalization. In addition to constantly optimizing and improving ERP and CRM, the company focuses on promoting the second phase of OA, terminal innovation, SRM, BI(Business intelligence), mobile Internet innovation and other projects.”

Similarly, we also define other types of digitalization related digital value chain, logistics, innovation, etc. Table B.3 displays the number of annual reports indicating different forms of

¹⁷This example also contains digital research and development

¹⁸This example also contains management digitalization.

digitalization. We see that the main forms of digitalization are production digitalization and management digitalization. The overall digitalization includes various types of digitalization we considered in our text analysis. The number of observations of overall digitalization is smaller than the treated observations as firms do not report their digitalization in all years after its initial implementation of digitalization.

Table B.3: Fields of Digital Transformation

Year	Overall	Production	Management	Value Chain	Logistics	Innovation
2010	33	19	15	0	2	6
2011	42	24	14	0	1	12
2012	54	38	21	0	2	19
2013	51	29	20	2	4	12
2014	105	75	40	2	6	26
2015	149	112	58	9	13	34
2016	221	164	80	18	27	45
2017	357	275	80	31	28	16
2018	461	341	116	43	46	35

Note: the definitions of all variables are explained in the data appendix.

B.2.2 Validation of the Measurement of Digital Transformation

The regression results in Table A.4 examine the impact of DT (DT_{it-1}) on firms' tangible and intangible assets. Two separate regression models are estimated, where the dependent variables are tangible assets and intangible assets, respectively. The findings highlight significant and differentiated effects of DT on these two types of assets.

The coefficient for DT_{it-1} is positive and statistically significant in both models, with a value of 0.028 ($p < 0.05$) for tangible assets and 0.060 ($p < 0.01$) for intangible assets. These results suggest that DT contributes to the growth of both tangible and intangible assets, with a notably stronger effect on intangible assets. This outcome aligns with the notion that DT is often associated with the enhancement of intangible resources, such as innovation capacity, intellectual property, and strategic repositioning, which are critical in the digital economy.

The results also underscore the importance of physical assets ($PP\&E_{it-1}$) and intangible assets ($Intangible_{it-1}$) as key predictors of their respective dependent variables. The coefficient for $PP\&E_{it-1}$ in the tangible assets model is 0.937 ($p < 0.01$), indicating strong persistence in the accumulation of physical assets. Similarly, the coefficient for $Intangible_{it-1}$ in the intangible assets model is 0.895 ($p < 0.01$), reflecting the enduring nature of intangible resources.

The plausibility of using annual reports to measure DT lies in their structured and regulated nature. Annual reports, mandated by the Securities Supervision Commission, provide firms with an opportunity to comprehensively outline their strategic priorities, including digital initiatives. While there is potential for firms to overstate or prematurely articulate their DT efforts

Table B.4: Responses in Tangible and Intangible Assets to Digital Transformation

	(1) Tangible Assets	(2) Intangible Assets
DT_{it-1}	0.028** (0.012)	0.060*** (0.016)
$PP\&E_{it-1}$	0.937*** (0.009)	
$Intangible_{it-1}$		0.895*** (0.007)
age	-0.113*** (0.014)	-0.106*** (0.014)
leverage	0.072* (0.040)	0.155*** (0.030)
duality	0.030*** (0.011)	0.005 (0.011)
largestshare	0.205*** (0.037)	0.196*** (0.041)
soe	0.290*** (0.062)	0.335*** (0.103)
constant	1.585*** (0.155)	2.234*** (0.137)
R^2	0.899	0.849

Note: The standard errors are in parentheses. ** $p < 5\%$, *** $p < 1\%$.

to attract investor attention, the keyword-based methodology used in this study helps address such concerns by focusing on specific textual contexts. By extracting and manually reviewing text surrounding digitalization-related keywords, the measure ensures that the analysis captures meaningful descriptions of firms' operational and strategic transformations rather than mere buzzwords. Furthermore, the manual coding process, where two independent coders evaluate the relevance of the extracted text, minimizes the risk of including superficial mentions of digitalization. This rigorous approach enhances the credibility of the measurement and its ability to reflect actual DT rather than rhetorical embellishments.

B.2.3 Productivity Estimation

We estimate the revenue-based total factor productivity by two-digit manufacturing industry. Limited by the sample size of some industries, we cluster some industry together to perform the productivity estimation. In particular, we re-group industries using the following procedures:

First, we maintain the industry classification for firms operating in special equipment manufacturing, chemical raw materials and chemical products manufacturing, computer, communication and other electronic equipment manufacturing, pharmaceutical manufacturing, general equipment manufacturing, electrical machinery and equipment manufacturing and instru-

ment manufacturing;

Second, we merge other two-digit manufacturing industries according to following rules: (1) Food manufacturing and agricultural and sideline food processing industries were merged with wine, beverage and refined tea manufacturing industries; (2) Chemical fiber manufacturing, rubber and plastic products, and petroleum processing, coking and nuclear fuel processing; Consolidation of ferrous metal smelting and rolling processing industry with non-ferrous metal smelting and rolling processing industry; (3) Consolidation of railway, marine, aerospace and other transport equipment manufacturing with automobile manufacturing; Consolidation of non-metallic mineral products, metal products and metal products, machinery and equipment repair; (4) Amalgamation of textile industry, textile clothing and apparel industry with leather, fur, feather and its products and footwear industry; (5) The industries of paper and paper products, printing and recording media reproduction were merged with the manufacturing industries of culture and education, arts and crafts, sports and recreation products; (6) Integration of furniture manufacturing with wood processing and wood, bamboo, rattan, palm and grass products; (7) Other manufacturing industries were merged with the industry comprehensive utilization of waste resources.

B.2.4 Propensity Score Model Estimation Results

Table B.5 reports the estimation results for the logit model used to estimate the propensity score. The dependent variable is the treatment indicator of digitalization, and the independent variables include firm age, leverage, duality, largest shareholding ratio, and state ownership. The results show that older firms, firms with higher leverage, firms with CEO duality, firms with a larger largest shareholder, and state-owned enterprises are more likely to undertake digital transformation. These results are consistent with our expectations and provide support for the validity of our propensity score estimation.

Table B.5: Logit Model for Propensity Score Estimation

Dependent Variable	digit	
age	0.918***	(0.073)
duality	0.152***	(0.055)
largestshare	0.433**	(0.177)
soe	0.156	(0.225)
leverage	0.186	(0.120)
constant	-4.699***	(0.219)
N	14280	

Note: The standard errors are in the parenthesis. ** $p < 5\%$, *** $p < 1\%$.

B.2.5 TWFE Results

We also estimate the productivity effects of DT using the conventional two-way fixed effects (TWFE) estimator. The results are reported in Table B.6. The results show that DT has a significant negative impact on total factor productivity (TFP), labor productivity, and capital efficiency, but a positive impact on total assets, number of employees, and total sales. These results are largely consistent with the results from the Callaway and Sant'Anna (2021) estimator.

Table B.6: Two-Way Fixed Effects Estimation Results

Dependent Variables	(1) TFP	(2) $\frac{\text{Sales}}{\text{Employees}}$	(3) $\frac{\text{Sales}}{\text{Fixed Assets}}$	(4) Assets	(5) Employees	(6) Sales
digit	-0.076*** (0.018)	-0.039** (0.018)	-0.055*** (0.011)	0.089*** (0.016)	0.115*** (0.018)	0.101*** (0.019)
age	0.065 (0.045)	-0.021 (0.043)	0.055* (0.028)	0.323*** (0.040)	0.464*** (0.044)	0.500*** (0.048)
leverage	-0.062 (0.039)	-0.129*** (0.036)	-0.210*** (0.024)	0.790*** (0.034)	0.846*** (0.037)	0.858*** (0.040)
duality	0.041*** (0.014)	-0.014 (0.014)	0.027*** (0.009)	0.005 (0.013)	0.025* (0.014)	0.008 (0.015)
largestshare	0.164** (0.071)	0.198*** (0.068)	-0.107** (0.044)	0.017 (0.063)	0.094 (0.069)	0.403*** (0.075)
soe	-0.025 (0.050)	0.335*** (0.049)	0.047 (0.031)	0.769*** (0.045)	0.363*** (0.049)	0.625*** (0.053)
constant	5.167*** (0.126)	13.620*** (0.120)	0.531*** (0.079)	20.543*** (0.111)	5.923*** (0.122)	19.239*** (0.132)
N	13222	14262	13235	14278	14268	14262

Note: The dependent variables are total factor productivity (TFP), labor productivity, capital efficiency, total assets, number of employees, and sales, respectively. The sample period is from 2011 to 2018. The standard errors are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

B.2.6 Falterers of Digital Transformation

In our analysis, we have focused on the initial adoption of digitalization strategies to examine the average treatment effect of DT on firm performance. However, we acknowledge that the success rate of digitalization is low, and some firms may falter in their digitalization efforts. To address the concern that mixing the falterers of DT with the non-falterers of digitalization may lead to misleading results, we have conducted a robustness check by separating the treated firms into falterers of DT and non-falterers of DT.

Table B.7 reports the estimation results for different scenarios of the continuity of DT. The results show that the negative impact of DT on productivity is more pronounced for non-falterers of digital transformers. The overall treatment effect is close to the estimation results for non-falterers of digital transformers. This suggests that consistent digital transformers are more likely to experience a negative impact on productivity compared to firms that falter in their digitalization efforts. Therefore, our main results are robust to the exclusion of firms that do not mention

digitalization in subsequent annual reports.

Table B.7: Productivity Effects of Digital Transformers: Falters vs. Non-Falters

	(1) All	(2) Non-Falters	(3) Falters
β^{agg}	-0.101*** (0.034)	-0.096** (0.039)	-0.088 (0.087)
Treated Firms	506	354	152
N	12412	11499	10234

Note: Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

B.3 Supplementary Robustness Checks

B.3.1 Measurements, Clustering, and Alternative estimators

Continuity of Digital Transformation We have defined the variable of DT in an absorbing manner. That is, our measure of digitalization emphasize the initial adoption of digitalization strategy, and does not distinguish consecutive digitalization strategies between non-consecutive ones. Note that it is not necessary that digitalization-related words appear in annual reports in a consecutive way: some firms claimed digitalization strategy in a certain year, but stopped in the following years. Also, other firms suspended digitalization for some years and resumed it afterwards. In the final estimation sample, of the treated 506 firms, 208 firms consecutively report digitalization in their annual reports, 146 firms report digitalization non-consecutively, while 152 firms are falters.

To address the concern that some firms may falter in their digitalization efforts, we conduct a robustness check by excluding firms that do not mention digitalization in subsequent annual reports. This allows us to examine the treatment effect of consistent digital transformers on firm performance. The results are reported in the appendix. The results show that the negative impact of DT on productivity is more pronounced for non-falters of digital transformers. The overall treatment effect is close to the estimation results for non-falters of digital transformers. This suggests that consistent digital transformers are more likely to experience a negative impact on productivity compared to firms that falter in their digitalization efforts. Therefore, our main results are robust to the exclusion of firms that do not mention digitalization in subsequent annual reports.

Content and Intensity of Digital Transformation We measure DT based on the annual reports of the publicly-traded firms. One concern is the validity of our measurement. We deal with this concern by considering two tests on the treatment effects of digitalization. First, we classify the digitalization by its content, i.e., production digitalization and management digi-

talization.¹⁹ We evaluate the treatment effects of these two different types of digitalization on various outcome variables. Second, we take into account the level of digitalization by considering the counts of words speaking to digitalization. For firms who mention digitalization more in the annual reports, we treat their digitalization as of higher intensity. If firms pay more attention to digitalization, we should expect that their impact on outcomes to be stronger.

Table B.8 show the estimation outcomes. The treatment effects of production digitalization on productivity is around -0.078, whereas -0.119 for management digitalization (Table B.8 *Panel* (a)). This shows that the finding of negative productivity impacts of digitalization is robust to the content of digitalization. We also find that production digitalization tend to increase total assets, total employees, and total sales more than management digitalization. This result is consistent with our understanding of production digitalization which usually introduces digitalized production technologies and requires a large amount of investment in hardware and software. *Panel* (b) in Table B.8 show that firms whose annual reports include more words related to digitalization experience more productivity drops and more gains in total assets, total employees, and total sales. Overall, these results indicate the plausibility of our text-based measure of DT and the robustness of our baseline results.

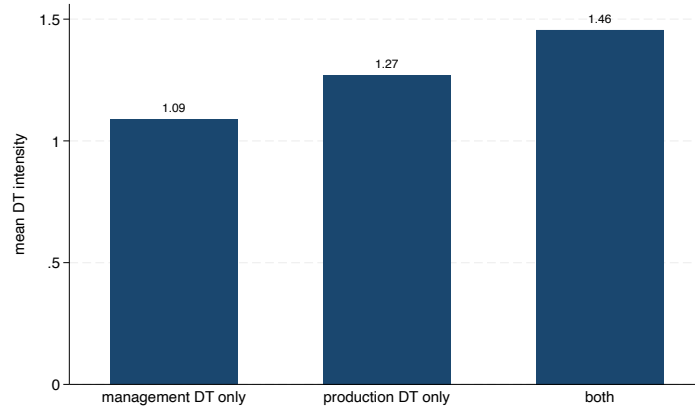
Table B.8: Treatment Effects by Types of Digitalization

	<i>(a) By Content</i>		<i>(b) By Intensity</i>	
	Production	Management	Low	High
TFP	-0.078** (0.038)	-0.119*** (0.043)	-0.087* (0.050)	-0.120** (0.056)
Total assets	0.087*** (0.029)	0.036 (0.036)	0.038 (0.041)	0.082** (0.034)
Total employees	0.108*** (0.029)	0.057 (0.036)	0.066* (0.036)	0.096*** (0.036)
Total Sales	0.048* (0.029)	-0.006 (0.041)	0.017 (0.040)	0.065* (0.037)
Treated Obs.	3275	1865	2360	2110

Note: All models include the firm's size, age, corporate governance variables (duality, largest shareholding, and SOE status), and industry dummies. The standard errors are obtained using Wild bootstrap with 1,000 draws and clustered at the firm level. ** $p < 5\%$, *** $p < 1\%$.

¹⁹We choose these two classifications mainly because of a lack of observations for other types of DT (See Table B.3).

Figure B.1: Mean Intensity of Digital Transformation by Content of Digitalization



Based on the results, one may infer that management digitalization is those with higher intensity of DT. To clarify this question, we report the mean intensity of DT by the content of DT in Figure B.1. As we can see from the figure, production digitalization contains more DT keywords than management digitalization, with a mean intensity of 1.27 compared to 1.09 for management digitalization. Furthermore, firms undergoing both production digitalization and management DT demonstrate the highest intensity of DT, with a mean value of 1.46. This suggests that these firms incorporate a broader range of digitalization efforts, reflected in the simultaneous presence of keywords related to both production and management digitalization in their annual reports.

Timing of Digital Transformation Another measurement issue may arise from the inaccurate timing of DT. We define firms to be treated units even since digitalization-related keywords appeared in their annual reports. However, if firms introduced DT earlier than the initial sample year (i.e., 2010), the timing of digitalization would be incorrect. One consequence is that the time of digitalization for the treated firms may be later than the actual year.²⁰ Both of these problems can bias the estimates of treatment effects. We use several methods to gauge the robustness of these results. First, note that firms starting DT in later periods have more pre-digitalization years for us to investigate the annual reports. We thus expect that the measurement error for their treatment timing would be less severe. *Panel (a)* in Table B.9 show that our baseline results remain robust when we restrict the sample to be containing only later treated units. Second, because the definition of never-treated units could be incorrect, we consider using all the not-yet treated units as controls as well.²¹ We find that the baseline estimates barely change (*Panel (b)*, Table B.9).

²⁰One may also be concerned that never-treated firms that should be classified as always treated units. However, we do not think this should be an issue as we have a panel of 9 years. Given that digitalization is a long process, it is rare that firms implemented digitalization before 2010 never mention it again afterwards.

²¹This triggers a different parallel trend assumption that requires the treated units and not-yet-treated units (including never treated ones) have a parallel trend. We empirically show that this parallel trend is satisfied.

Table B.9: Robustness Checks: Later Timing, Clustering, and Alternative Controls

<i>(a) Treatment Effects with Later Timing:</i>				
Outcomes	TFP	Total Employees	Total Assets	Total Sales
$\beta^{agg}(e \geq 2012)$	-0.057** (0.028)	0.075*** (0.024)	0.060** (0.023)	0.053** (0.026)
$\beta^{agg}(e \geq 2013)$	-0.066*** (0.021)	0.050** (0.022)	0.037* (0.022)	0.022 (0.026)
$\beta^{agg}(e \geq 2014)$	-0.065*** (0.021)	0.047** (0.022)	0.051** (0.023)	0.033 (0.027)
<i>(c) Not-yet Treated in Control Groups:</i>				
Outcomes	TFP	Total Employees	Total Assets	Total Sales
β^{agg}	-0.098*** (0.035)	0.059** (0.029)	0.077*** (0.027)	0.042 (0.029)
<i>(b) Standard Errors Clustered at City Level:</i>				
Outcomes	TFP	Total Employees	Total Assets	Total Sales
β^{agg}	-0.081** (0.037)	0.055** (0.024)	0.049* (0.027)	0.030 (0.034)

Note: All models include the firm's size, age, corporate governance variables (duality, largest shareholding, and SOE status), and industry dummies. The standard errors are obtained using Wild bootstrap with 1,000 draws and clustered at the firm level (except for panel (c)). ** $p < 5\%$, *** $p < 1\%$.

Level of Clustered Standard Errors In our baseline results, standard errors are clustered at firm level. *Panel (c)* in Table B.9 show the estimation results when clustering the standard errors at the city level. This means that our estimation results are robust to potential spatial and time-series correlations in the idiosyncratic errors.

B.3.2 Alternative Staggered DID Estimators

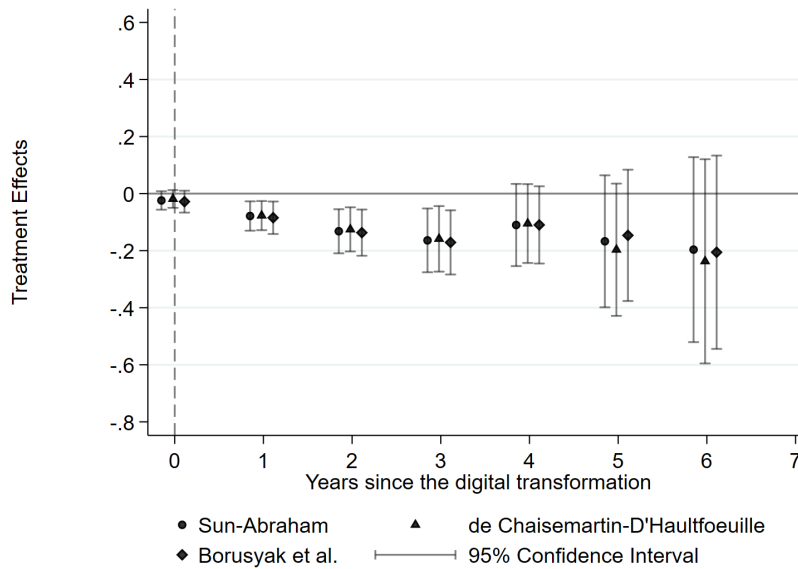
We consider several alternative staggered DID estimators to gauge the robustness of our baseline results. First, we consider the Callaway and Sant'Anna (2021) estimator using not-yet treated as controls. Second, we employ alternative estimators introduced by Sun and Abraham (2021), de Chaisemartin and d'Haultfoeuille (2020), and Borusyak et al. (2021) to estimate the treatment effects.

Callaway and Sant'Anna (2021) Estimator with Not-yet Treated Units in Controls Considering that our sample has a relatively large groups of never-treated firms, we do not use not-yet treated firms as controls in the baseline estimation. One may be wondering how adding the not-yet treated observations may alter the estimation results. Including the not-yet treated units as controls triggers a stronger parallel trends assumption on the pretrends across different treatment groups. We perform tests and find that parallel trends assumption is satisfied. *Panel (a)* in Table B.9 shows the aggregate treatment effects when using not-yet treated as con-

trols. Overall, the estimated treatment effects stay very close to the baseline results. Therefore our results are robust to using not-yet-treated as controls.

Other Estimators In our baseline analysis, we employ the Callaway and Sant’Anna (2021) estimator and only use never treated units as the controls. One important advantage of this estimator is the parallel trends in relatively weak in the sense that it does not restrict pre-trends across different treatment groups of units. As a robustness check, we also consider alternative staggered DID estimators. As discussed in Marcus and Sant’Anna (2021) and Borusyak et al. (2021), these estimators make different assumptions on the parallel trends and employ different units in the control group. First, de Chaisemartin and d’Haultfoeuille (2020), Sun and Abraham (2021), and Borusyak et al. (2021) impose a parallel pretrends condition across all treatment groups. As a result, it allows the researcher to employ any not-yet-digitalized firms by year t as a valid comparison group for those firms already implemented DT by year t . Second, compared to de Chaisemartin and d’Haultfoeuille (2020) and Sun and Abraham (2021), Borusyak et al. (2021) use all non-treated units to impute the potential outcome for the treated units, thus can potentially increase the efficiency.

Figure B.2: Treatment Effects on TFP: Alternative Estimators



Note: All the parallel pre-trends are satisfied and related results are omitted to save space.

Figure B.2 shows estimates using three alternative estimators. We find that the point estimates and 95% confidence intervals are quite similar for all three estimators, and are very close to the baseline results reported in Figure 1.

B.3.3 More on Endogeneity Issues: Instrumental Strategy

Firms' DT decisions may be driven by unobserved firm-level strategies, such as concurrent investments in product R&D or green technologies that also affect productivity, which potentially violates our parallel-trends assumption. To purge such confounding, we turn to a leave-one-out, sectoral shift-share instrument à la Bartik (1991). For firm i in industry j and year t , define:

$$\overline{DT}_{it} = \frac{\sum_{k \neq i, k \in j} DT_{k,t}}{N_{j,t} - 1} \quad (\text{B.1})$$

where $DT_{k,t}$ denotes the explanatory variable of firm i belonging to the same industry j as firm i , and $N_{j,t}$ is the number of firms in industry j at time t . By construction, this instrument captures exogenous, sector-wide DT momentum-weighted by peer firms-while excluding firm i 's own digital choices.

This instrument strategy addresses several major endogeneity concerns. First, it is relatively independent of firm-specific shocks. Any idiosyncratic push by firm i to adopt non-digital strategies (e.g., an R&D spree or a green-tech pilot) does not mechanically move $\overline{DT}_{it}$, which depends only on other firms' DT. Second, our firm fixed effects soak up time-invariant confounders including sectoral common trends (e.g., sector-wide policy shifts, talent pools) that might simultaneously drive DT and productivity. Third, it is likely that peers' DT intensity affects firm i 's productivity through competitive pressures to digitalize, but not via direct knowledge spillovers nor resource transfers for two main reasons: (1) Input markets in the manufacturing sector are geographically and technologically dispersed, making direct knowledge spillovers less likely; (2) The DT process is often firm-specific, requiring tailored investments in technology and processes that are not easily transferable across firms.

We recognize that no instrument is perfectly insulated from every conceivable confounder, and so $\overline{DT}_{it}$ could in principle pick up other unobserved, time-varying firm strategies.²² However, these violations arise only under very special circumstances: (1) if a industry-wide regulation forced all firms in industry j both to digitalize and undertake large-scale green investments in the same period, peers' DT would proxy for environmental spending; (2) if a dominant conglomerate spun off units and imposed a digitalization mandate exclusively within its original industry, peers' DT would reflect hidden restructuring motives; (3) if a sudden, unobserved liquidity crunch hit industry j , delaying peers' digital projects while forcing firm i to postpone other strategic investments, correlation could arise. In practice, environmental mandates are staggered, mega-reorganizations are well-publicized and absorbed by our firm-by-year fixed effects, and China's diversified corporate banking system makes sector-specific credit freezes both unlikely and transient. Outside these exceptional scenarios, peers' digitalization intensity provides a clean, exogenous source of variation; coupled with our rich controls and fixed effects, any remaining bias from these worrying events is likely negligible.

²²We thank the referee for highlighting this potential limitation.

Table B.10: First-stage regressions

Dependent Variable:	<i>Digital</i>	
	(1)	(2)
<i>Rival-DT</i>	0.705*** (0.087)	0.725*** (0.088)
<i>Controls</i>	No	Yes
Firm FEs	Yes	Yes
Year FEs	Yes	Yes
<i>N</i>	13342	13222
R^2	0.660	0.663
F statistics	66.083	13.12

Note: Control variables include the firm's size, age, and corporate governance variables (duality, largest shareholding, and SOE status). The standard errors are in parentheses. ** $p < 5\%$, *** $p < 1\%$.

Table B.10 columns (1) and (2) show the first-stage estimation results. The rival firms' digitalization is strongly correlated with the firm's own DT whether we add controls or not. The R^2 is around 0.66, indicating the proportion of variance of the potential endogenous variable explained by the explanatory variables of the model is relatively high. The F-statistic is greater than 10, strongly supporting the strength of the instrumental variables.

Table B.11 reports the second-stage regression results. Columns (1) and (2) consistently show that digitalization has a significant negative impact on total factor productivity. The Klerbergen-Paap rk statistic for the under-identification strongly rejects that the model is under identified, and the Klerbergen-Paap F statistic is larger the rule-of-thumb value of 10, ruling out that the instrument is a weak instrument. In Columns (3) and (4), we also display the results for OLS estimator for the two-way fixed effects model. Because the two-way fixed effects estimator is potentially biased, we should interpret the results with caution. However, given that (1) the consistency between the results of IV regressions and OLS regressions; and (2) the consistency between OLS estimate and our baseline estimate, we believe that our results is not likely to be driven by the endogeneity problem caused by time-varying unobserved firm-level factors.

Table B.11: IV and OLS Regressions on Digitalization and Productivity

Dependent Variable	Total Factor Productivity			
	IV		OLS	
	(1)	(2)	(3)	(4)
<i>Digital</i>	-2.080*** (0.416)	-1.890*** (0.375)	-0.072*** (0.026)	-0.089*** (0.026)
<i>Age</i>		-0.034 (0.116)		0.009 (0.075)
<i>Leverage</i>		-0.155 (0.102)		-0.214** (0.090)
<i>Duality</i>		0.083** (0.034)		0.040 (0.026)
<i>Largest Share</i>		0.401** (0.201)		0.176 (0.167)
<i>SOE</i>		-0.241** (0.115)		-0.148* (0.082)
<i>ln(Assets)</i>		0.227*** (0.036)		0.178*** (0.029)
Firm FEs	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes
<i>N</i>	13342	13222	13342	13222
<i>R</i> ²			0.954	0.956
K-P LM statistic	54.007 [0.000]	55.531 [0.000]		
K-P F statistic	66.08	68.44		

Note: K-P LM statistic is the LM version of Kleibergen-Paap *rk* statistic (Kleibergen and Paap, 2006) for under-identification test, with P values reported in square brackets. The K-P F statistic is the Wald F version of Kleibergen-Paap *rk* statistic (Kleibergen and Paap, 2006) for the test on weak identification. The standard errors in parentheses are obtained by clustering at firm level. * $p < 10\%$, ** $p < 5\%$, *** $p < 1\%$.